

PACT: Persona-Aware Coordinated Thompson Sampling for LLM Multi-Agent Systems

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Abstract

LLM agents are increasingly deployed in negotiation, market design, and public-good provision, where a coordinator must orchestrate agents whose private dispositions (altruist, free-rider, cautious negotiator) silently determine welfare and fairness yet are never directly observed. Tracking these dispositions reduces to inference in a heterogeneous-type Markov game whose joint posterior is exponential in the agent count and whose only feedback is the agents’ behaviour and outcomes. We introduce **PACT** and **PACT**⁺, posterior-sampling algorithms that maintain a per-agent persona belief, sample a joint world per episode, and plan in it. Personas are realised as system-prompt templates and a calibrated persona-conditioned observation model supplies the likelihood for an outside-the-prompt Bayes update. The ⁺ variant adds a type-discrimination bonus that steers planning toward states where candidate personas diverge.

Under persona-independent transitions, per-agent reward locality, and prior factorisation, the per-agent marginals equal the joint posterior, giving $O(n|\Theta_i|)$ rather than $O(|\Theta_i|^n)$ storage, an $\tilde{O}(\sqrt{K})$ Bayesian regret bound, and an $\Omega(|\Theta_i|^n)$ burn-in improvement from the bonus on a deep-trap family. Without prior factorisation no algorithm achieves better than $\Omega(\sqrt{|\Theta_i|^n K})$, so the structural assumption set is necessary for sublinear-in- n regret. Across four LLM backbones and three coordination benchmarks, PACT⁺ cuts cumulative regret by 6 to 12× over the strongest published LLM-coordination baselines where the preconditions hold, stays within 1% of a strict focal-payoff ceiling on every compact Concordia Pub Coordination configuration, and remains within 0.1 of the per-backbone leader on SOTOPIA-Hard, an open-ended dialogue transfer whose curated dyads, LLM-judge reward, and heuristic persona surrogate fall outside the exact theory. The em-

pirical results therefore separate theorem-aligned evidence on HP-SPGG from descriptive transfer under increasing substrate mismatch.

1 Introduction

LLMs are increasingly deployed as multi-agent systems for cooperative AI, negotiation, mechanism design, and economic simulation (Zhang et al., 2021; Leibo et al., 2017; Perolat et al., 2017). In any realistic group the agents are not interchangeable, since each carries a private *persona* (a private utility, style, or goal) that fixes its rewards and remains hidden from the coordinator. Ignoring this heterogeneity costs regret linear in the horizon, while maintaining a single belief over the joint persona profile requires a table exponential in the agent count.

PACT (Algorithm 1) maintains one small belief per agent over its persona and one belief over transitions. Each episode it draws a joint guess from these beliefs, plans a coordinated policy that is optimal for that guess, executes the policy through the LLM agents, and updates each belief from a persona-conditioned likelihood of the observed action or outcome. *PACT*⁺ (Algorithm 2) augments planning with a *type-discrimination bonus*, a shaping term that prefers states at which candidate personas would behave differently. The bonus requires no additional LLM calls and decays as beliefs concentrate.

The joint posterior over personas is exponential in n , yet under three explicit conditions (persona-independent transitions, per-agent reward locality, and prior factorisation across agents) the product of per-agent marginals equals the joint posterior at every episode. These conditions are restrictive deployment requirements, not properties that can be inferred from benchmark performance. They also yield an $\tilde{O}(\sqrt{K})$ Bayesian regret bound, and for the bonus an exponential-in- n reduction of burn-in regret on a deep-trap family. We defer formal statements and proofs to Appendix A.2 through Appendix A.5.2, and give operational assumption checks in §4.

Our contributions are fourfold. First, we introduce two LLM-substrate algorithms, PACT and

PACT⁺, that require no fine-tuning and support either token-level action likelihoods or calibrated outcome likelihoods. Second, we construct a controlled benchmark, HP-SPGG, as a diagnostic substrate that satisfies all six structural assumptions by construction, on which type-aware coordination beats ECON (Xie et al., 2025) and A-ToM (Mu et al., 2026) by 6 to 12 $\times$ at $K=20$ on four backbones (DeepSeek-V3.2, GPT-5.4-nano, Kimi-K2.6, Llama-4-Maverick-17B-128E-Instruct-FP8). Third, we transfer to two independent substrates: compact Concordia (Vezhnevets et al., 2023), where PACT⁺ remains within 1% of the strict focal ceiling on every Pub Coordination configuration, and SOTOPIA-Hard (Zhou et al., 2024), a dyadic open-text benchmark on which the substrate-adapted surrogate remains competitive overall and is examined in detail on three exploratory private-goal families. Fourth, we develop a regret analysis combining (a) a posterior-decoupling theorem under TI+RL+PF, (b) an $\Omega(\sqrt{|\Theta_i|^n K})$ lower bound when PF is dropped that establishes PF as the structural condition enabling sublinear-in- n dependence, and (c) an $\Omega(|\Theta_i|^n)$ burn-in separation between vanilla and bonus-augmented PACT on a deep-trap family.

2 Related Work

LLM-derived Bayesian belief tracking. Recent work has explored using LLMs as components of Bayesian belief tracking, in three distinct ways. *LLM-generated natural-language hypotheses:* Hypothetical Minds (Cross et al., 2025) scaffolds multi-agent Theory of Mind in Melting Pot by generating, evaluating, and refining natural-language hypotheses about opponent strategies. Arumugam and Griffiths (Arumugam and Griffiths, 2026) implement PSRL for LLM agents in single-agent MDPs: a textual posterior describes uncertainty about the MDP, one LLM samples a plausible MDP hypothesis from that text, a second LLM acts optimally under the sampled hypothesis, and a third LLM rewrites the textual posterior after observing a trajectory. Thus their posterior state and posterior update are themselves natural-language objects. *Symbolic Bayesian-style reasoning:* CoBel-World (Wang et al., 2025) maintains a symbolic belief world and performs zero-shot Bayesian-style updates through LLM reasoning for embodied multi-agent collaboration, and Li et al. (2023b) maintain an explicit belief state in cooperative text games. *LLM-derived likelihoods with closed-form Bayes update:* LAIP (Gelpi et al., 2025) uses LLM-simulated action likelihoods for single-target inverse planning with mathematical posterior updates. Amin (2026) treat LLMs as approximate likelihood models for cost-aware single-agent classification with multi-LLM ensemble aggregation. Mu et al. (2026) and CSP4SDG (Xu et al., 2025) apply closed-form poste-

rior updates over persona / role priors in single-user adaptive querying and social deduction respectively. PACT belongs to the third camp but extends it to the multi-agent coordination Markov game setting. Three structural conditions (TI, RL, PF) admit a per-agent posterior factorisation theorem (Theorem A.8) that yields $O(n|\Theta_i|)$ posterior storage versus $O(|\Theta_i|^n)$ for any non-factored joint coordinator. The factored Bayesian update is paired with a backward-induction CCE planner for coordinated multi-agent action selection and admits a $\tilde{O}(\sqrt{K})$ Bayesian regret bound (Theorem A.12). The structural decoupling result and the heterogeneous-type Markov game regret bound are the loci of novelty; the LLM-as-likelihood primitive itself is a shared technical ingredient across this line of work.

Posterior-sampling reinforcement learning (Strens, 2000; Osband et al., 2013; Agrawal and Jia, 2017; Ouyang et al., 2017) attains near-optimal Bayesian regret in single-agent MDPs, with the information-theoretic framework of Russo and Van Roy (2014, 2016) underpinning our type-inference analysis. In Markov games, posterior sampling has so far been restricted to zero-sum settings, including Jafarnia-Jahromi et al. (2024) for average-reward and Xiong et al. (2022) with function approximation. All assume known type structure.

On latent-type and heterogeneous MARL, Kwon et al. (2021) prove that learning latent MDPs is exponentially hard without identifiability, and Nguyen-Tang and Arora (2024) extend hardness to adaptive adversaries. Albrecht and Ramamoorthy (2013); Albrecht et al. (2019) introduce Harsanyi-Bellman Ad-hoc Coordination, a Bayesian opponent-type method with asymptotic convergence but no finite-time regret. Our decoupling theorem and $\tilde{O}(\sqrt{K})$ bound provide the finite-sample complement.

For general-sum equilibrium learning, a line on V-learning and its variants establishes sample complexity for CCE in general-sum MGs (Jin et al., 2021; Song et al., 2022; Mao and Başar, 2023; Erez et al., 2023; Yang et al., 2025a), and the PSRO family (Lanctot et al., 2017; Muller et al., 2020; Marris et al., 2021) targets equilibrium of an empirical game via population learning. Neither line maintains a Bayesian posterior over latent payoff structure, so our framework is complementary, and PSRO-style oracles can be plugged into the inner CCE step of PACT without affecting the decoupling result.

Structured Bayesian RL is closer in spirit. Mutti et al. (2024) factor dynamics via causal priors, and Arumugam and Griffiths (2026) revive PSRL for LLM agents by assigning LLMs the PSRL subroutines of posterior update, posterior sampling, and sampled-model policy execution. Neither factors the prior across hidden agent types, and neither

gives a multi-agent posterior decoupling result.

General-purpose LLM multi-agent frameworks. AutoGen (Wu et al., 2023), CAMEL (Li et al., 2023a), AgentVerse (Chen et al., 2023), and MetaGPT (Hong et al., 2023) provide conversation protocols, role assignment, and orchestration infrastructure. They can host a PACT coordinator, but do not by themselves define a hidden-type posterior, an observation likelihood, or a policy for repeated social-dilemma coordination. We therefore position them as complementary systems frameworks rather than interchangeable algorithmic baselines; ECON and A-ToM are the closer comparisons because they explicitly reason about beliefs or partners under strategic interaction.

3 PACT: Persona-Aware Coordinated Thompson Sampling

3.1 Setting

We formalise centralised LLM coordination as an episodic, finite-horizon, heterogeneous-type Markov game (HT-MG)

$$\mathcal{M} = (\mathcal{N}, \mathcal{S}, \{\mathcal{A}_i\}_{i \in \mathcal{N}}, P^*, \{r_i\}_{i \in \mathcal{N}}, H, s_1, \{\Theta_i\}_{i \in \mathcal{N}}, P_0),$$

with $\mathcal{N} = \{1, \dots, n\}$ agents, finite state space $\mathcal{S}$, per-agent finite action space $\mathcal{A}_i$, joint action space $\mathcal{A} = \prod_{i \in \mathcal{N}} \mathcal{A}_i$ with elements $a = (a_1, \dots, a_n)$, transition kernel $P^* : \mathcal{S} \times \mathcal{A} \rightarrow \Delta(\mathcal{S})$, horizon $H \in \mathbb{N}$, and fixed initial state s_1 .

Each agent carries a private *persona* $\theta_i^* \in \Theta_i$ drawn from a finite archetype set Θ_i (e.g. *altruistic builder*, *free rider*); we write $|\Theta_i| = m$ when archetype sets are uniform, $\theta^* = (\theta_1^*, \dots, \theta_n^*) \in \Theta := \prod_{i \in \mathcal{N}} \Theta_i$ for the joint persona profile, and $\theta^* \sim P_0^\Theta$ for its prior. Per-agent reward $r_i : \mathcal{S} \times \mathcal{A} \times \Theta_i \rightarrow [0, 1]$ depends on the agent’s own persona only. The coordinator’s prior over the unknown environment factors as $P_0 = P_0^P \otimes P_0^\Theta$, with $P^* \sim P_0^P$ on the transition kernel and P_0^Θ on the joint persona.

Interaction protocol. The coordinator interacts with the n LLM agents over K episodes, each of length H . At the start of episode $k \in \{1, \dots, K\}$, the system resets to $s_1^k = s_1$. At every turn $h \in \{1, \dots, H\}$ the coordinator observes the joint state s_h^k , selects a joint action $a_h^k \in \mathcal{A}$ (executed through the LLM agents under their respective persona prompts), observes per-agent rewards $r_h^k = (r_{i,h}^k)_{i \in \mathcal{N}}$ with $r_{i,h}^k = r_i(s_h^k, a_h^k, \theta_i^*)$, and transitions to $s_{h+1}^k \sim P^*(\cdot | s_h^k, a_h^k)$. The episode trajectory is $\tau_k = (s_h^k, a_h^k, r_h^k)_{h=1}^H$, and the history available to the coordinator at the start of episode k is $\mathcal{F}_k = \sigma(\tau_1, \dots, \tau_{k-1})$. The personas θ^* and the kernel P^* are fixed across all K episodes and never directly observed.

Policies, value, equilibrium. A coordinated (non-stationary) policy is a map $\pi = (\pi_h)_{h=1}^H$ with $\pi_h : \mathcal{S} \rightarrow \Delta(\mathcal{A})$. The per-agent value under π in a model (P, θ) is

$$V_{i,h}^\pi(s; P, \theta) = \mathbb{E} \left[\sum_{h'=h}^H r_i(s_{h'}, a_{h'}, \theta_i) \mid s_h = s, \pi, P \right],$$

and the joint social value is $V_h^\pi(s; P, \theta) = \sum_{i \in \mathcal{N}} V_{i,h}^\pi(s; P, \theta)$. A coarse-correlated-equilibrium (CCE) oracle returns

$$\pi^*(P, \theta) \in \arg \max_{\pi \in \text{CCE}(P, \theta)} V_1^\pi(s_1; P, \theta),$$

which we assume deterministic and measurable in (P, θ) (used throughout the analysis). The benchmark policy is $\pi^* := \pi^*(P^*, \theta^*)$, the policy that would be optimal under full knowledge of the kernel and the persona profile.

Theory–implementation boundary for planning. The regret theorem treats the planner as an ϵ_{CCE} -approximate CCE oracle. The reported finite HP-SPGG and compact Concordia experiments instead enumerate every joint action profile and take a deterministic welfare- or focal-objective argmax; this is an exact centralised planner for the reported finite objective, but it is not an implementation of a general CCE linear program. SOTOPIA-Hard has neither an enumerable action set nor a CCE step: utterances are generated autoregressively by the LLM policy. We report candidate-limited planning sensitivity below so tracker quality is not conflated with search quality.

Posteriors and per-agent marginals. At episode k the coordinator maintains a posterior over the unknowns,

$$\mu_k(P, \theta) := \mathbb{P}(P^* = P, \theta^* = \theta \mid \mathcal{F}_k),$$

together with its transition marginal $\mu_k^P(P) = \sum_{\theta} \mu_k(P, \theta)$ and the per-agent persona marginals $\mu_k^{\Theta_i, i}(\theta_i) = \sum_{\theta_{-i}, P} \mu_k(P, \theta_{-i}, \theta_i)$, where $\theta_{-i} \in \prod_{j \neq i} \Theta_j$ denotes the personas of the remaining agents. The naive joint-posterior representation has size $|\Theta| = \prod_i |\Theta_i|$, exponential in n ; PACT (§3.2) maintains only the marginals $\{\mu_k^{\Theta_i, i}\}_{i \in \mathcal{N}}$ at total storage $O(n|\Theta_i|)$.

Bayesian regret. Performance is measured by the cumulative *Bayesian regret* of a learning algorithm $\mathcal{L}$ producing the sequence of executed policies $\{\pi_k\}_{k=1}^K$:

$$\mathfrak{R}_K(\mathcal{L}) := \mathbb{E} \left[\sum_{k=1}^K \left(V_1^{\pi^*}(s_1; P^*, \theta^*) - V_1^{\pi_k}(s_1; P^*, \theta^*) \right) \right], \quad (1)$$

where the expectation is over $(P^*, \theta^*) \sim P_0$, the trajectories $\{\tau_k\}$, and any internal randomness of

$\mathcal{L}$ (including the per-episode samples in PACT). The full game tuple, the six structural assumptions (transition independence, reward locality, prior factorisation, type identifiability, and two regularity conditions used in the analysis), and the symbols introduced here are summarised in Appendix A.1.

3.2 PACT: one belief per agent

Maintaining one belief over the full persona profile requires a table of size $\prod_i |\Theta_i|$, exponential in the number of agents. PACT (*Persona-Aware Coordinated Thompson sampling*) instead keeps one independent belief per agent, $\mu_k^{\Theta,i}$, over its persona, plus one belief μ_k^P over transitions, for total storage $O(n|\Theta_i|)$. Three mild structural conditions (Appendix A.1) make this exact rather than approximate.

Assumption 3.1 (structural conditions for decoupling, informal). *(TI) The transition kernel P^* is independent of the persona profile θ^* . (RL) Each agent’s reward is parametrised by its own persona, $r_i(s, a, \theta_i^*)$, although the joint action $a = (a_i, a_{-i})$ and state s may freely couple agents. (PF) The prior factors across agents, $P_0(\theta^*) = \prod_i P_0^{\Theta,i}(\theta_i^*)$.*

(TI) says the physics of the world depends only on actions, not on who the agents are. (RL) says the persona dependence is private even though actions still couple agents through the state. (PF) says the coordinator’s prior treats agents as a priori independent. RL fails for utilities of the form $r_1(s, a, \theta) = r_1^{\theta_1}(s, a) - \alpha r_2^{\theta_2}(s, a)$ (envy, reciprocity, fairness preferences), since the agent-2 persona enters agent-1’s reward; Appendix A.4 shows the decoupling theorem breaks in this regime. HP-SPGG and the compact mechanistic Concordia abstraction evaluated in Section 4.2 satisfy all three by construction. SOTOPIA-Hard is not an isolated PF stress: its curated dyads do not expose a factorised generative prior, its LLM evaluator does not guarantee RL, and its open-text actions and heuristic likelihood surrogate lie outside the tabular known-model setting.

Theorem 3.2 (Posterior decoupling). *Under Assumption 3.1, the joint posterior over (P^*, θ^*) factors exactly as the product of the transition posterior and the per-agent persona marginals at every episode k :*

$$\mathbb{P}(P^*, \theta^* | \mathcal{F}_k) = \mu_k^P(P^*) \prod_{i=1}^n \mu_k^{\Theta,i}(\theta_i^*), \quad (2)$$

where $\mathcal{F}_k$ is the history through episode k . The per-agent factored update of PACT is therefore statistically equivalent to maintaining the full joint posterior. (Proof: Appendix A.2.)

The theorem certifies that the cheap factored representation is not an approximation. The product of n tables of size $|\Theta_i|$ is bit-for-bit equal to the

Algorithm 1 PACT

Input: priors $P_0^P, \{P_0^{\Theta,i}\}$; horizon H ; episodes K ; CCE oracle.

- 1: Initialise $\mu_1^P \leftarrow P_0^P, \mu_1^{\Theta,i} \leftarrow P_0^{\Theta,i}$ for all i .
 - 2: **for** $k = 1, \dots, K$ **do**
 - 3: Sample $\hat{P}_k \sim \mu_k^P$ and $\hat{\theta}_i^k \sim \mu_k^{\Theta,i}$ independently.
 - 4: $\pi_k \leftarrow \text{CCE}(\hat{P}_k, \hat{\theta}_k)$; execute π_k , observe τ_k .
 - 5: Bayes-update each marginal independently from τ_k .
 - 6: **end for**
-

joint table the naive coordinator would compute. Storage drops from $O(|\Theta_i|^n)$ to $O(n|\Theta_i|)$ at no statistical cost, and any guarantee that holds for the joint update transfers to the factored update.

Concretely, PACT’s belief state is a set of numeric tables, not text in a prompt. For agent i , the row $\mu_k^{\Theta,i}(\theta_i)$ stores the coordinator’s probability that agent i has persona θ_i . Let $x_{i,h}^k$ collect the variables available before the inferred observation (public history for an action-token channel; public history plus executed joint action for an outcome channel), and let $o_{i,h}^k$ be that observation. After the episode, the coordinator evaluates each observation under every candidate persona,

$$\begin{aligned} \ell_{i,h}^k(\theta_i) &= q_\phi(o_{i,h}^k | x_{i,h}^k, \theta_i), \\ L_{i,k}(\theta_i) &= \prod_{h=1}^H \ell_{i,h}^k(\theta_i), \\ \mu_{k+1}^{\Theta,i}(\theta_i) &= \frac{\mu_k^{\Theta,i}(\theta_i) L_{i,k}(\theta_i)}{\sum_{\theta'_i \in \Theta_i} \mu_k^{\Theta,i}(\theta'_i) L_{i,k}(\theta'_i)}. \end{aligned} \quad (3)$$

The LLM never writes or normalises the posterior. When an API exposes stable token logprobs, q_ϕ may be the persona-prompted action probability; fixed verbalizers, top- k truncation, and decoding calibration then become systems assumptions. In the reported HP-SPGG runs, however, q_ϕ is a Gaussian outcome likelihood centred on a cached LLM-judge calibration tensor ($\sigma=0.08$), so online token-logprob access and a verbalizer are not required. If neither a calibrated outcome model nor token probabilities are available, Eq. (3) cannot be interpreted as calibrated Bayes. The coordinator multiplies the supplied likelihoods, normalises the table, and samples from the resulting marginals; the factorisation theorem concerns this exact update under TI+RL+PF, irrespective of which calibrated observation channel instantiates q_ϕ .

The implementation uses three primitives. First, every candidate persona $\theta_i \in \Theta_i$ is realised as a system-prompt block (in HP-SPGG, *altruistic builder*, *conditional cooperator*, *strategic hedger*, *free rider*). Second, a finite-model planner selects a joint action without an LLM call. Third,

a persona-conditioned observation model supplies q_ϕ in Eq. (3). The transition belief μ_k^P is updated by Dirichlet conjugacy on observed (s, a, s') transitions. Only the third primitive is substrate-dependent: HP-SPGG uses cached calibrated outcomes, a logprob-enabled deployment may use action tokens, and SOTOPIA uses an explicitly uncalibrated heuristic instead of exact Bayes.

3.3 PACT⁺: a small bonus that breaks the deep-trap

If the policy recommended by the sampled world drives the agents into a region where every candidate persona predicts the same reward, the coordinator observes nothing that separates personas and per-agent beliefs stay uniform. The coordinated policy ties up, the rollout remains in the no-information region, and the coordinator can stay stuck for an exponential-in- n number of episodes before reaching a discriminating state. Appendix B formalises this as the *deep-trap* family.

PACT⁺ adds, during the backward-induction step, a small reward at each (s, a) where two candidate personas disagree,

$$D_k(s, a) := \sum_{i=1}^n \mathbb{E}_{\theta, \theta' \sim \mu_k^{\Theta, i}} [r_i(s, a, \theta) - r_i(s, a, \theta')]. \quad (4)$$

$D_k(s, a)$ is large when the coordinator is still uncertain about an agent’s persona and two candidate personas would visibly disagree at (s, a) . The planner adds $\beta D_k(s, a)$ ($\beta = 0.25$ throughout) to the Q-function during the same backward-induction pass that produces the coordinated policy. As beliefs sharpen, $D_k \rightarrow 0$ on its own and the bonus disappears, leaving the asymptotic rate unaffected.

In the formal algorithm, D_k is recomputed once at the start of episode k from μ_k , evaluated for all (s, a) during that episode’s backward-induction pass, and held fixed until the next posterior update; a system that updates beliefs within an episode should instead recompute it after each update. The finite HP-SPGG code uses the monotone proxy $\tilde{D}_k(a) = \sum_i (1 - \max_\theta \mu_k^i(\theta)) \text{Var}_\theta[r_i(a, \theta)] + 0.05 \text{sd}(a)$ and posterior-expected welfare for the reported PACT⁺ row. Thus the theorem applies to Eq. (4), while the empirical ⁺ row is a transparent model-based adapter rather than a literal implementation of Algorithm 2.

PACT⁺ therefore changes only the planning objective, not the belief machinery. The numeric posterior, the factorised Bayes update in Eq. (3), and the LLM interface are identical to PACT; the extra term simply asks the planner to prefer actions that are useful for distinguishing still-plausible personas when doing so is cheap.

The bonus is computed from the per-agent beliefs at $O(n|\Theta_i|^2)$ per (s, a) and slots into the same backward-induction pass as PACT, so only policy

Algorithm 2 PACT⁺

Input: as in Alg. 1, plus $\beta > 0$.

- 1: Initialise as in Alg. 1.
 - 2: **for** $k = 1, \dots, K$ **do**
 - 3: Sample $\hat{P}_k, \hat{\theta}_i^k$ as in Alg. 1; compute D_k via Eq. (4).
 - 4: Solve CCE with $Q_h^+(s, a) += \beta D_k(s, a)$ in backward induction.
 - 5: Execute, observe τ_k , Bayes-update marginals.
 - 6: **end for**
-

execution and per-token persona scoring call the LLM. We fix $\beta = 0.25$ across every backbone and substrate, and it is never re-tuned. The bonus is *representation-aware*, since it needs per-agent marginals that the joint posterior recovers only by paying $O(|\Theta_i|^n)$ marginalisation cost; it is *self-decaying*, because $D_k \rightarrow 0$ as $\mu_k^{\Theta, i}$ concentrates, so the bonus stops perturbing the policy once beliefs sharpen; and it is *tie-breaking*, in the sense that it places strictly positive mass on persona-discriminating states. With $\beta = 0$, PACT⁺ reduces bit-identically to PACT.

Theorem 3.3 (Burn-in separation, informal). *On the deep-trap family $\mathcal{G}_m^{\text{trap}}$ parameterised by per-agent persona-set size m and agent count n , the expected burn-in regret of vanilla PACT, and of any algorithm restricted to the joint posterior, is $\Theta(m^n H)$, whereas PACT⁺ with any bonus weight $\beta > 0$ achieves $O(H)$. The gap is $\Omega(m^n)$, exponential in the agent count. (Formal statement and proof: Appendix B.)*

Without the bonus, the coordinator can remain in the no-information region for $\Theta(m^n)$ episodes, paying $\Theta(H)$ regret per episode, and doubling the agent count squares the wait. Any positive β steers the planner toward persona-disagreeing states, the coordinator gathers discriminating evidence within $O(1)$ episodes, and burn-in regret collapses to $O(H)$. The separation is obtained without any change to the LLM interface, any extra LLM call, or any new hyperparameter beyond the fixed $\beta = 0.25$. The $\Omega(m^n)$ lower bound on vanilla PACT rests on a deterministic CCE tie-break (Assumption CSR in Appendix A.1); randomised tie-breaking weakens it by a problem-dependent constant in $(0, 1]$, leaving the qualitative separation intact but not necessarily the exact rate.

Three formal guarantees follow (proofs in Appendix A.2 through Appendix A.5.2). (P1) Under TI+RL+PF, PACT attains $\tilde{O}(\sqrt{K})$ Bayesian regret at $O(n|\Theta_i|)$ storage, near-tight against the minimax lower bound and exponentially tighter than the bound that holds when PF is dropped (Theorem A.16). (P2) On an explicit exploration-trap instance, PACT is sublinear while MAP-Type-Greedy

Table 1: Implemented HP-SPGG algorithms. The formal analysis permits a CCE oracle; these finite experiments use deterministic exhaustive joint-profile argmax.

Name	Belief representation	Action selection
PACT	per-agent marginals $\{\mu^{\theta_i}\}$	posterior sample $\rightarrow$ exhaustive argmax
PACT ⁺	per-agent marginals	posterior mean $+\beta D_k \rightarrow$ exhaustive argmax
Joint-PSRL	full joint μ^θ	posterior sample $\rightarrow$ exhaustive argmax
MAP-Type-Greedy	per-agent marginals	type argmax $\rightarrow$ exhaustive argmax
PSRL-NoType	uniform	uniform sample $\rightarrow$ exhaustive argmax

and PSRL-NoType suffer linear regret. (P3) On the deep-trap family parameterised by m and n , the bonus cuts burn-in regret from $\Theta(m^n H)$ to $O(H)$ for any $\beta > 0$. The empirical β -sweep of Figure 2 tests P3; P1 and P2 are corroborated by per-axis scaling and posterior-concentration diagnostics in Appendix C.1.

PACT extends classical PSRL (Strens, 2000; Osband et al., 2013) from single-agent MDPs to Markov games with private agent personas. The closest LLM-agent analogue is the LLM-PSRL implementation of Arumugam and Griffiths (2026). In plain terms, their method asks LLMs to carry out vanilla PSRL in language: maintain a textual posterior over an unknown MDP, sample one textual MDP hypothesis, act under that hypothesis, and rewrite the textual posterior from the observed trajectory. PACT differs in three ways. First, the latent variable is not a whole single-agent MDP but the vector of hidden agent personas in a general-sum Markov game. Second, the posterior is not a natural-language summary rewritten by an LLM; it is an explicit numeric posterior updated outside the prompt by Bayes’ rule from persona-conditioned LLM likelihoods. Third, under TI+RL+PF this numeric posterior factorises across agents, converting $O(|\Theta_i|^n)$ joint-type storage into $O(n|\Theta_i|)$ without statistical penalty and making the type-discrimination bonus D_k computable in $O(n|\Theta_i|^2)$ rather than $O(|\Theta_i|^n)$. The bonus then cures the deep-trap burn-in failure of vanilla PSRL on a Joint-PSRL implementation, which under matched RNG is otherwise pathwise-identical to PACT (Proposition A.10).

4 LLM Experiments

We evaluate on three substrates ordered by distance from the formal model. HP-SPGG is a controlled diagnostic substrate that we construct to satisfy all six HT-MG-BC assumptions by construction; its role is to test the predicted regret behaviour and the β -sweep theory, not to claim broad applicability. We then transfer to the compact public Concordia games from the NeurIPS 2024 Cooperative AI Contest, externally designed by DeepMind, and finally to SOTOPIA-Hard as an out-of-model stress test with curated dyads, open-ended language actions, LLM-judge reward, and a heuristic four-

type surrogate. All runs use four LLM backbones spanning vendors, capacities, and weight access, namely DeepSeek-V3.2 (open-weight MoE), GPT-5.4-nano-20260317 (small closed), Kimi-K2.6 (mid open), and Llama-4-Maverick-17B-128E-Instruct-FP8 (large open).

Deployment diagnostics and stop rules. Before invoking the theorem, practitioners should audit: **TI**, by testing whether next-state residuals retain persona information conditional on (s, a) ; **RL**, by varying partner type while holding (s, a, θ_i) fixed and checking whether focal reward changes; **PF**, by estimating cross-agent prior dependence rather than imposing independent marginals; **TID/calibration**, by reporting held-out likelihood, reliability, and minimum pairwise separation; **library coverage**, by posterior-predictive checks and an out-of-library likelihood threshold; and **planner quality**, by the exact-versus-approximate objective gap. Failure of TI or RL calls for a joint type-dynamics/reward model, failure of PF calls for a joint prior, and failure of likelihood or coverage checks makes PACT a heuristic. Performance on SOTOPIA cannot substitute for these tests.

Metric. The coordination metric is cumulative *regret*, defined as the per-episode shortfall against the policy that would be optimal under full persona knowledge, and reported across the four backbones on every substrate. Per-axis scaling diagnostics (E1 posterior concentration, E2 $|\Theta_i|$ -scaling, E3 n -scaling, E5 per-episode trajectories), which test the three predictions (P1) through (P3) of §3.3, are given in Appendix C.1.

4.1 HP-SPGG: Controlled Diagnostic Benchmark

Setup. HP-SPGG is constructed by us as a diagnostic substrate. A coordinator oversees $n=3$ player LLMs across $K=20$ rounds. Each player’s hidden type $\theta_i \in \Theta_i$ is drawn independently with replacement from a uniform prior over $|\Theta_i|=4$ persona archetypes constructed in the SOTOPIA style (Zhou et al., 2024) and spanning the empirical taxonomy of Fischbacher et al. (2001) (altruistic builder, conditional cooperator, strategic hedger, free rider). The implementation has $|\mathcal{A}_i|=5$ contribution levels, hence $5^3=125$ joint profiles at the headline cell. The joint persona posterior has $|\Theta_i|^n=64$ profiles, while PACT’s factored representation occupies $4 \cdot 3=12$ entries. TI, RL, PF, ISI, and CSR hold by construction; the retained calibration diagnostic reports a minimum pairwise mean absolute reward-profile gap ≥ 0.05 . We use this as an empirical separation check, not as a direct Hellinger estimate. The benchmark therefore tests predicted behaviour in a controlled setting; Concordia tests static-geometry transfer and SOTOPIA only descriptive out-of-model transfer.

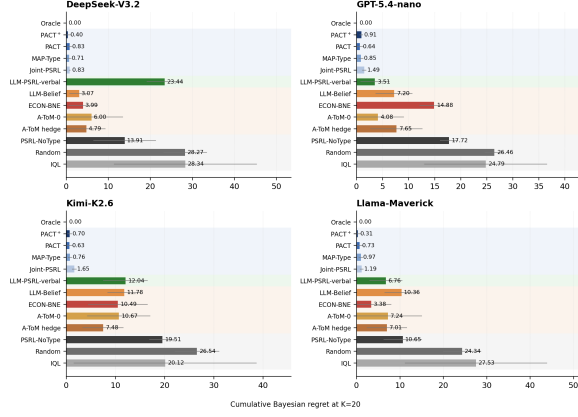


Figure 1: HP-SPGG cumulative Bayesian regret at $K=20$ across four LLM backbones (5 seeds per cell, calibration c19; bar heights are means and each bar is numerically labelled). Four families: *Bayesian type-aware* (blue), *LLM-PSRL* (green), *LLM-coordination* (warm), and *type-agnostic* (gray). LLM-PSRL implements posterior sampling with natural-language hypothesis sampling and posterior rewriting, adapting Arumugam and Griffiths (2026) to multi-agent coordination. The PACT family attains the lowest non-oracle regret on every backbone.

Baselines. We compare PACT⁺ ($\beta=0.25$, fixed across backbones) against three families. The *Bayesian type-aware* family covers PACT, Joint-PSRL, and MAP-Type-Greedy. The *LLM-coordination* family covers ECON-BNE (Xie et al., 2025), A-ToM at orders $\{0, 1, 2\}$ with Hedge/FTL aggregation (Mu et al., 2026; Cesa-Bianchi et al., 1997), and the prompted `llm_belief` and `llm_greedy`. The *type-agnostic* family covers PSRL-NoType, IQL, and Random.

Figure 1 reports the cumulative regret of each baseline on the four backbones. The PACT family attains 0.40 on DeepSeek, 0.64 on GPT-5.4-nano, 0.63 on Kimi, and 0.31 on Llama-Maverick, the lowest non-oracle regret on each, with separations of $7.7\times$, $6.3\times$, $11.8\times$, and $10.8\times$ over the best LLM-coordination baseline and 27 to $35\times$ over the best type-agnostic baseline. The LLM-PSRL baseline attains 23.44 on DeepSeek, 3.51 on GPT-5.4-nano, 12.04 on Kimi-K2.6, and 6.76 on Llama-Maverick at the same $n=3$, $K=20$ cell, between $10.9\times$ and $24.8\times$ worse than PACT. This is the empirical cost of replacing PACT’s closed-form numeric update with a natural-language posterior representation. The identity of the second-best baseline rotates across backbones (`llm_belief` on DeepSeek, A-ToM-0 on GPT-5.4-nano, A-ToM hedge on Kimi, ECON-BNE on Llama-Maverick), indicating that prompt-based belief reasoning is sensitive to backbone capacity.

The three clusters group by algorithm family, not by backbone, and on Llama-Maverick the PACT

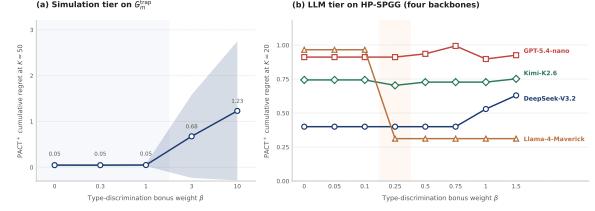


Figure 2: β -sweep of PACT⁺ on HP-SPGG c19 across four backbones. Three regimes predicted by Theorem A.15 appear: *extreme self-decay* (GPT, Kimi); *self-decay edge* (DeepSeek); *burn-in jump* (Llama-Maverick, $3\times$ improvement at $\beta=0.25$).

family reaches 0.31 while type-agnostic methods sit at 8 to 11. The second-best baseline rotates from cell to cell, yet PACT⁺ is first on every backbone, with the largest margins ($11.8\times$ and $10.8\times$) on the two backbones where prompt-based belief reasoning is shakiest. Moving belief-tracking out of the prompt and into a closed-form numeric update removes a compounding source of reasoning error: prompt-based baselines can misreason about their own posterior across $K=20$ episodes, whereas PACT deterministically applies the cached persona-conditioned outcome likelihoods.

Figure 2 sweeps $\beta \in \{0, 0.05, 0.1, 0.25, 0.5, 0.75, 1.0, 1.5\}$ and reveals three distinct regimes, each predicted by the theorem. On Llama-Maverick the bonus produces a sharp $3\times$ drop from 0.97 to 0.31 at $\beta=0.25$, the exponential burn-in improvement of Theorem A.15(ii). On DeepSeek the regret is flat across $\beta \in [0, 0.75]$ and then rises, the high- β boundary of (i). On GPT-5.4-nano and Kimi-K2.6 the curves are essentially flat, because the posterior concentrates faster than the bonus has time to add useful exploration. Per-cell calibration robustness across all 92 HP-SPGG cells, together with four scaling experiments (E1, E2, E3, E5) realising the $\Omega(|\Theta_i|^n)$ -vs- $O(1)$ envelope and the $\tilde{O}(\sqrt{K})$ asymptote of Theorems A.15 and A.14, are consolidated in Appendix C.1 (Figure 5).

Which regime a backbone lands in is determined by how fast its persona-conditioned outcome likelihoods concentrate the posterior in early episodes. GPT-5.4-nano and Kimi-K2.6 yield sharply separated calibrated reward surfaces by $k=1-2$, so D_k collapses before the bonus can point the planner at any state, and the curve is flat because nothing remains for the bonus to fix. Llama-Maverick is the opposite extreme: its calibrated surface is diffuse enough at low k that vanilla PACT visits the deep-trap on a non-trivial fraction of seeds, and the bonus gets it out. DeepSeek is intermediate, since concentration is fast enough that small β is harmless but $\beta > 0.75$ over-perturbs the planner and lifts regret. A single $\beta = 0.25$ therefore works

across all four backbones (it never hurts more than 0.04 regret and rescues the worst backbone by $3\times$), which is why we hold it fixed.

Table 2 separates tracker and planner failure. Controlled top- k , temperature, and zero-mean log-likelihood perturbations leave regret below 0.005, but fixed calibration drift of 0.05 raises regret to 0.73, and restricting the planner to 32 candidates raises it to 1.41. The 50% persona mixture is an important warning: regret remains at zero while posterior mass on the nominal type falls to 0.37, so task success is not evidence that the persona library is semantically correct. The full 17-condition sweep is in Appendix C.3. This experiment characterises the calibrated HP-SPGG observation model; it does not estimate SOTOPIA projection accuracy.

4.2 Concordia Public Benchmarks

To rule out benchmark bias we evaluate on the public Concordia substrates (Vezhnevets et al., 2023; Smith et al., 2025), independently designed by DeepMind. For each substrate we map agent disposition to θ_i , exogenous state to s , and per-agent decisions to a_i , and Concordia provides analytic reward via the substrate’s coordination payoff component, eliminating the LLM judge. We compare PACT⁺ (joint proxy and per-agent MAP variants) against ECON-BNE-mech and A-ToM-1-mech adapted to the Concordia interface. The oracle reference (`oracle_focal`, dashed in Figure 3) is the deterministic focal-payoff argmax over $\prod_i \mathcal{A}_i$ via `itertools.product` under full one-hot persona information, the analog of HP-SPGG’s analytic argmax for the reported focal metric; see Appendix C.4 for the unified construction across all three LLM benchmarks.

Structural mapping and scope. The qualification *compact* is important. Each reported seed is collapsed to a one-shot static case. In Pub Coordination, favorite pubs, the relationship matrix, and venue closure are fixed at case construction; the exact payoff then depends on the full joint pub choice. Another player’s preference therefore affects focal reward through that player’s public action, which RL permits, and there is no persona-conditioned next-state kernel, so TI holds vacuously. The official `fruitville_gullible` and `vegbrooke_stubborn` Haggling configs do define fixed supporting-player responses in the full dialogue simulator. Our compact runner, however, does not execute those NPC policies: it reads static role counts, valuation/cost ranges, and action menus, then centrally chooses both proposal and acceptance. Thus the reported variants stress static payoff/action geometry rather than TI. PF is imposed by the independent per-player proxy marginals rather than tested from trajectories. Finally, the mechanistic rows use the exact known Concordia payoff and full case infor-

mation; their proxy posteriors are constructed from relationship statements or deal spreads, not from an unknown transition/reward model. The separate live-LLM axis relaxes information access, but does not make the mechanistic panel an unknown-model benchmark.

Figure 3 reports focal-score means across 7 Pub Coordination configurations spanning four village variants and three closure regimes, and 9 Haggling sub-configurations (fruitville and vegbrooke variants, single and multi-item). On Pub Coordination, PACT⁺ (joint proxy) stays within 1% of `oracle_focal` on every mechanistic config, while ECON-BNE-mech trails by 5 to 15% and A-ToM-1-mech by 5 to 30%. On Haggling, `oracle_focal` is the focal-payoff ceiling; PACT⁺ wins or ties among non-oracle baselines on all 9 configs while remaining below the privileged focal oracle in asymmetric-role cases where the focal objective can extract more surplus. On `vegbrooke_strange`, A-ToM-1-mech collapses to the substrate floor at -10.0 while the mechanistic joint proxy holds at 0, a known-model planning gap rather than evidence of hidden-NPC inference.

The substrates are externally designed by DeepMind and rewards are analytic (no LLM judge). Within the compact abstraction, closure regime, relationship topology, valuation range, role composition, and item multiplicity change the static payoff/action geometry while preserving TI and RL. PACT⁺ stays within 1% of `oracle_focal` on every Pub Coordination config, showing transfer across these externally chosen geometries rather than across hidden transition dynamics. In Haggling, the focal oracle deliberately maximises the focal agent’s own payoff and may therefore exploit asymmetric roles more aggressively than the Nash-fair PACT⁺ policy. The A-ToM-1-mech collapse on `vegbrooke_strange`, where its lowball rule selects a substrate-penalised action, illustrates planner robustness in this known-model compact game; it is not evidence about recovery from a latent NPC transition process.

4.3 SOTOPIA-Hard: Partial Transfer Beyond Theoretical Preconditions

SOTOPIA-Hard (Zhou et al., 2024) is a dyadic ($n = 2$) social-negotiation benchmark with open-ended natural-language actions. Its 70 cases are curated environment–profile pairs rather than draws from an explicit persona prior; the per-episode reward is a seven-dimensional LLM-judge score conditioned on both agents’ goals and the joint transcript. The substrate therefore lies outside HT-MG-BC on more than PF: finite/tabular actions fail, PF is not identified, RL is not guaranteed by the joint-goal evaluator, and neither a known reward model nor a calibrated persona-conditioned action likelihood is available. We test descriptive transfer using a

Table 2: Controlled HP-SPGG deployment robustness ($K=20$, 500 paired seeds, $n=3$, $|\Theta_i|=4$, $\beta=0.25$; no new LLM calls). Regret is evaluated in the unperturbed environment against exhaustive full-information argmax. Temperature, log-noise, and top- k perturb only the likelihood update; drift perturbs the candidate outcome model; mix interpolates adjacent true personas while inference retains the original library; candidate-32 restricts planning to 32 of 125 joint profiles. Lower regret and higher true-type mass are better.

Condition	Cumulative regret	True-type mass	Diagnostic interpretation
Reference	0.002 ± 0.000	0.995	Calibrated outcome likelihood, exhaustive planner
Likelihood temperature $T=4$	0.004 ± 0.001	0.860	Over-dispersed but decision-stable
Log-likelihood noise $\sigma_\ell=1$	0.005 ± 0.002	0.873	MAP accuracy falls to 0.882
Top-2 likelihoods only	0.002 ± 0.000	0.995	No degradation in this separated finite library
Calibration drift $\sigma_c=0.05$	0.730 ± 0.036	0.699	Systematic model error is the dominant inference risk
50% adjacent-persona mixture	0.000 ± 0.000	0.371	Decisions remain good while persona semantics fail
32-profile candidate planner	1.410 ± 0.027	0.995	Search error dominates despite accurate tracking

substrate-adapted four-class intent/persona surrogate in Sotopia self-play. The intended adapter initializes a numeric table uniformly and applies hand-specified keyword increments after partner utterances, but a post-hoc implementation audit found that the archived 70-case runs read the agent inbox while SOTOPIA 0.1.5 places the previous action in `Observation.last_turn`. Those retained runs therefore made zero numeric-table updates and must be interpreted as prompt-conditioned transfer with a uniform surrogate state, not as recurrent Bayes updating. The update path was corrected before the separate menu-corruption sensitivity experiment reported in Appendix C.5.

Let ϕ denote this project-defined map from a full profile or utterance to the four surrogate classes. SOTOPIA supplies neither native labels for those classes nor an intent confusion matrix, so projection error $\Pr[\phi(x) \neq z]$ and its effect on the LLM judge are not identifiable from the retained data; we therefore do not report “projection accuracy.” The profile-informed `oracle_belief` averages 3.103, slightly below the best non-oracle average (3.110), and is itself exceeded by A-ToM-1 on Llama (3.359 vs. 3.282). This is direct evidence that the projection is not a ground-truth persona oracle. The corrected menu-corruption experiment measures intervention sensitivity rather than accuracy; a deployment still requires a held-out labelled sample and predictive calibration under ϕ before treating the surrogate state as persona inference.

In the corrected GPT-5.4-nano corruption run (120 episodes per $p \in \{0, .1, .2, .3\}$), the $p=0$ self-play means are 2.665, 3.229, and 2.743 on *craigslist*, *donate*, and *revenge*, below the retained GPT-specific four-baseline references (2.761, 3.386, 3.200); the operational advantage-disappearance threshold is therefore $p^*=0$ because no initial advantage exists.

Paired changes across nonzero p are non-monotone and comparable to their descriptive uncertainty, while the provider exposes no sampling seed, so the experiment supports a sensitivity bound but no causal dose-response claim (Appendix C.5).

Aggregated across all 11 SOTOPIA-Hard families, no method dominates. The per-backbone leader rotates and the five baselines cluster within a 0.12-score envelope (Appendix C.5, Table 9). After first averaging each baseline across backbones and family instances, PACT⁺ leads by +0.06, +0.10, and +0.02 on the three exploratory families in Figure 4; cell-wise, however, its mean margin against the best alternative is negative on every family, and no implementation-level test assigns these three to an assumption-satisfying class. On the remaining eight families PACT⁺ stays within 0.33 of the per-cell leader. The dashed `oracle_policy` reference sits ~ 0.43 above every non-oracle baseline, but it combines privileged access to a profile-derived surrogate class with best-of- $K=5$ search under the same judge. The much smaller non-oracle spread therefore shows that action search and evaluator variation dominate differences among the retained belief prompts; it does not isolate PF or validate the theorem on this substrate.

Figure 4 exposes a qualitative trajectory difference that the end-of-dialogue bars hide. On *craigslist_bargains* and *donate_funds*, several prompt baselines peak at turns $k=3-4$ and then decline by $k=6$, while the PACT⁺ row finishes at or above its mid-dialogue score. On *revenge_plot* its judged score rises from 1.65 to 2.93. Because the post-hoc audit found zero numeric-table updates in these archived runs, this pattern cannot be attributed to continued belief revision. It instead combines method-specific prompting, autoregressive generation, dialogue history, and repeated

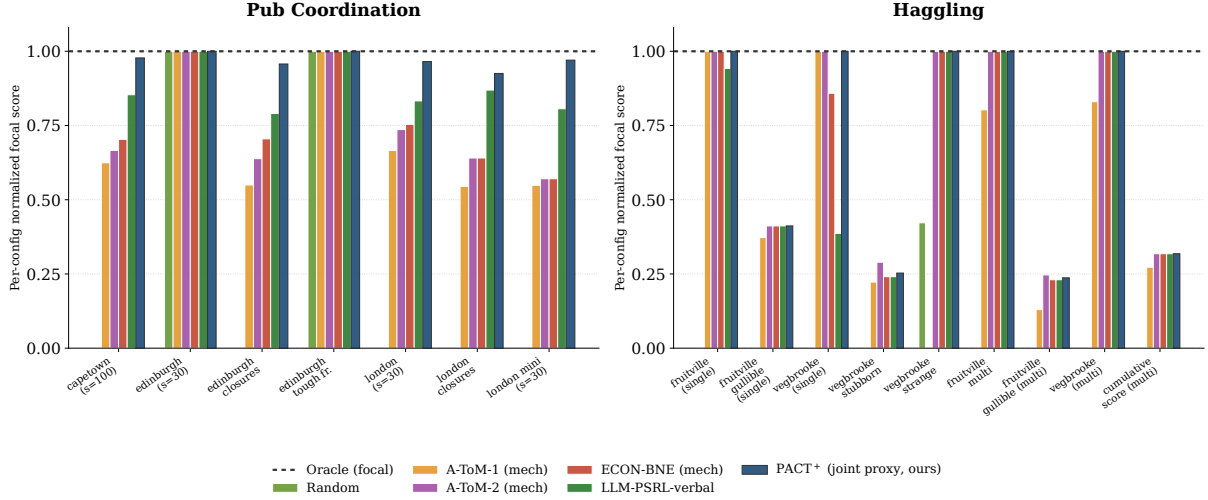


Figure 3: Concordia results, per-axis min-max normalised so the best plotted method reaches the outer ring. *Left*: Pub Coordination, 9 axes (7 village configs at $s=30$, one at $s=100$, one DeepSeek-V3.2 live LLM rollout at $s=5$). *Right*: Hagglng, 9 axes spanning 5 single-item and 4 multi-item configs at $s=30$. The filled blue polygon is PACT⁺; the dashed `oracle_focal` line is a strict upper bound for the plotted focal metric, computed by full-information enumeration—not the legacy joint-welfare reference. Absolute Pub scores are PACT⁺ 1.21–1.32 versus focal ceiling 1.23–1.34 (maximum gap 0.029); Hagglng focal scores span 0–7.82, with non-oracle margins 0–0.25.

LLM judging. We therefore retain the curves only as descriptive dialogue diagnostics; the corrected menu-corruption experiment is the relevant tracker-sensitivity check.

5 Discussion, Limitations, and Conclusion

We introduced PACT and PACT⁺, two posterior-sampling algorithms for centralised multi-agent LLM coordination under unknown personas and dynamics. The factored posterior is exact under three structural conditions, gives exponential storage savings in n , and matches a near-tight $\tilde{O}(\sqrt{K})$ Bayesian regret bound that becomes $\Omega(\sqrt{|\Theta_i|^n K})$ as a lower bound when prior factorisation is dropped, establishing PF as the structural condition that enables sublinear-in- n dependence. The type-discrimination bonus reduces burn-in regret by an $\Omega(|\Theta_i|^n)$ factor on the formal deep-trap family. The strongest theorem-aligned evidence is HP-SPGG; compact Concordia shows transfer across externally defined static payoff geometries. On SOTOPIA-Hard, the substrate-adapted heuristic is competitive but does not dominate, and selected-family gains are exploratory rather than a validation of TI, RL, or PF.

The guarantees assume tabular state/action spaces, a fixed discrete persona library, a calibrated observation model, and the six stated structural conditions; extending the proof to function approximation or continuous/learned personas is an open problem rather than an automatic corollary of PSRL.

Personas are sampled once per episode and held fixed, so non-stationary personas are out of scope. Token-likelihood deployment requires stable candidate logprobs and verbalizers, but the reported HP-SPGG implementation instead uses cached outcome likelihoods; Table 2 shows that zero-mean likelihood perturbations are benign in this controlled domain while systematic calibration drift is not. A low regret under persona mixtures does not validate the library: at 50% interpolation the nominal true-type mass is only 0.37. SOTOPIA’s keyword surrogate receives no theorem guarantee.

Cost and operational trade-off. The complete research sweep is estimated at roughly 5×10^5 calls and 6.5×10^8 tokens (Appendix D.1); this is an experiment-suite total, not a per-deployment requirement. The reported HP-SPGG policy evaluations reuse a one-time calibration tensor and add no online LLM calls, while an online token-likelihood implementation costs up to $nH|\Theta_i|$ scoring queries per episode in addition to action generation. PACT⁺ adds no calls over PACT. Because ECON, A-ToM, prompted baselines, mechanistic Concordia, and cached PACT use different call structures, a single “calls per percent improvement” number would be misleading; we instead report the per-interface formula, one-time calibration cost, and wall-clock totals separately. Candidate-limited planning results show that saving CPU search, even with a correct posterior, can cost more regret than noisy likelihoods.

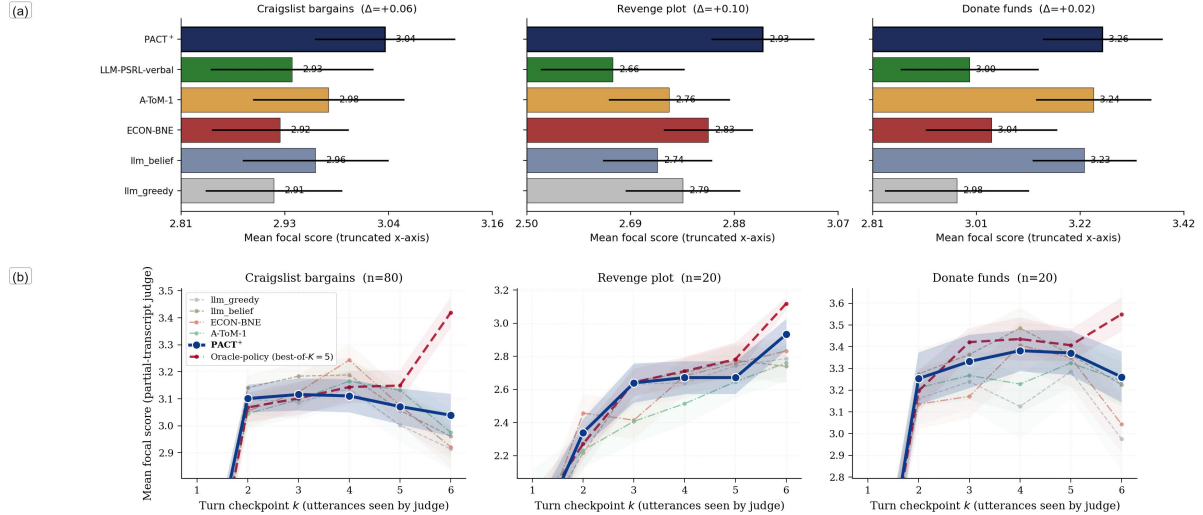


Figure 4: SOTOPIA-Hard, three exploratory private-goal scenario families selected for detailed analysis: *craigslislt_bargains* ($n=80$), *revenge_plot* ($n=20$), and *donate_funds* ($n=20$). (a) End-of-dialogue focal score, including LLM-PSRL as a natural-language belief baseline. (b) Per-turn trajectory of the same LLM-judge score for the original baseline panel; the red dashed line is *oracle_policy*, a profile-informed best-of- $K=5$ search reference with extra generation/evaluation budget, not a strict oracle (Appendix C.4). The remaining eight families are reported in aggregate in Appendix C.5; the 3/8 split is descriptive, not a verified TI/RL/PF classification.

Misuse and fairness. Inferring a discrete persona can invite stereotyping or differential treatment in applications such as dispatch and ride-hailing. The posterior should be used only for the stated coordination objective, audited for subgroup calibration and outcome disparities, and equipped with an out-of-library/abstention path. It should not be interpreted as a stable psychological attribute, especially when reward locality fails or the library is hand-authored.

Decentralised HT-MG-BC (each agent maintains its own posterior over the others), beyond-TI dynamics (persona-coupled transitions), function-approximation extensions to continuous state-action spaces, and adaptive- β schedules informed by posterior-concentration estimates are natural next steps.

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A Theoretical Supplement

A.1 HT-MG-BC Formal Setting

Heterogeneous-Type Markov Game. An HT-MG is a tuple $\mathcal{G} = \langle \mathcal{S}, \{\mathcal{A}_i\}, P^*, \{\Theta_i\}, \{r_i\}, P_0, H \rangle$ with finite state space $|\mathcal{S}| = S$, joint action space $\mathcal{A} = \prod_i \mathcal{A}_i$ of size A_{joint} , type space $\Theta = \prod_i \Theta_i$, transition kernel $P^*: \mathcal{S} \times \mathcal{A} \times \Theta \rightarrow \Delta(\mathcal{S})$, per-agent reward $r_i: \mathcal{S} \times \mathcal{A} \times \Theta_i \rightarrow [0, 1]$, joint prior P_0 , and horizon H . At the start of each episode a type profile $\theta^* \in \Theta$ is sampled once and held fixed; P^* is fixed across episodes but unknown. A centralised learner observes the full trajectory τ_k and updates a posterior over (P^*, θ^*) .

Structural assumptions. The decoupling theorem rests on three: **(TI)**, Assumption A.1 the transition kernel does not depend on the type profile; **(RL)**, Assumption A.2 each r_i depends only on θ_i (other agents' actions a_{-i} may still influence reward); **(PF)**, Assumption A.3 the prior factorises as $P_0(P, \theta) = P_0^P(P) \prod_i P_0^{\Theta, i}(\theta_i)$. The regret analysis additionally uses type-identifiability **(TID)**, Assumption A.4: per-step Hellinger gap $\geq \rho$, initial-state independence **(ISI)**, Assumption A.5, and a deterministic CCE selection rule **(CSR)**, Assumption A.6. Both TI and RL are individually necessary: violating either breaks the likelihood factorisation, as shown by two counter-examples in Appendix A.4.

Assumption A.1 (TI). $P^*(s' | s, a, \theta) = P^*(s' | s, a)$.

Assumption A.2 (RL). $r_i(s, a, \theta) = r_i(s, a, \theta_i)$.

Assumption A.3 (PF). $P_0(P, \theta) = P_0^P(P) \prod_i P_0^{\Theta, i}(\theta_i)$.

Assumption A.4 (TID). $\inf_{s, a} D_H^2(P(\cdot | s, a, \theta_i), P(\cdot | s, a, \theta'_i)) \geq \rho > 0$ for all $\theta_i \neq \theta'_i$.

Assumption A.5 (ISI). $s_1^k \sim \beta$ fixed and independent of (P^*, θ^*) .

Assumption A.6 (CSR). $\text{CCE}(\cdot, \cdot)$ is a measurable deterministic mapping.

Performance criterion. Let $\pi^{\text{CCE}}(P, \theta)$ denote a CCE policy of the MG induced by (P, θ) . The type-conditional Bayesian regret of agent i over K episodes is

$$\text{Reg}_i^B(K) := \mathbb{E}_{(P^*, \theta^*) \sim P_0} \sum_{k=1}^K \left(V_i^{\pi^{\text{CCE}}(P^*, \theta^*)}(s_1^k, \theta_i^*) - V_i^{\pi_k}(s_1^k, \theta_i^*) \right). \quad (5)$$

A.2 Posterior Decoupling Theorem

This section states and proves the main structural result.

A.2.1 Likelihood Factorization

We first derive the conditional likelihood of an observed trajectory given the latent world. Under PACT, the policy π_k at episode k is a function only of the history $\mathcal{H}_{k-1}$ and the algorithmic posterior samples $(\hat{P}_k, \hat{\theta}_k)$, which are themselves measurable with respect to $\mathcal{H}_{k-1}$. Crucially, π_k does not depend on the true latent (P^*, θ^*) .

Lemma A.7 (Joint Likelihood Factorization). *Under Assumptions A.1 and A.2, the trajectory likelihood at episode k factorizes as*

$$P(\tau_k | P, \theta, \mathcal{H}_{k-1}) = c_k(\tau_k) \cdot L_P(\tau_k; P) \cdot \prod_{i \in [n]} L_{\Theta, i}(\tau_k; \theta_i) \quad (6)$$

where

$$c_k(\tau_k) := \mathbb{E}_{\pi_k | \mathcal{H}_{k-1}} \left[\prod_{h=1}^H \pi_k(a_h^k | s_h^k) \right], \quad (7)$$

$$L_P(\tau_k; P) := \prod_{h=1}^H P(s_{h+1}^k | s_h^k, a_h^k), \quad (8)$$

$$L_{\Theta, i}(\tau_k; \theta_i) := \prod_{h=1}^H P(r_h^{k, i} | s_h^k, a_h^k, \theta_i). \quad (9)$$

The constant $c_k(\tau_k)$ depends on τ_k and $\mathcal{H}_{k-1}$ but not on (P, θ) .

Proof. Conditional on π_k , the trajectory likelihood unrolls along the time axis as

$$P(\tau_k | \pi_k, P, \theta) = \prod_{h=1}^H \pi_k(a_h^k | s_h^k) \cdot P(s_{h+1}^k | s_h^k, a_h^k, \theta) \cdot \prod_{i \in [n]} P(r_h^{k, i} | s_h^k, a_h^k, \theta_i). \quad (10)$$

Apply Assumption A.1 to drop θ from the transition factor, and Assumption A.2 to localize the reward factor to θ_i :

$$P(\tau_k | \pi_k, P, \theta) = \prod_{h=1}^H \pi_k(a_h^k | s_h^k) \cdot \prod_{h=1}^H P(s_{h+1}^k | s_h^k, a_h^k) \cdot \prod_{h=1}^H \prod_{i \in [n]} P(r_h^{k, i} | s_h^k, a_h^k, \theta_i). \quad (11)$$

Marginalize over π_k with respect to $P(\pi_k | \mathcal{H}_{k-1})$, which is independent of (P, θ) :

$$\begin{aligned} P(\tau_k | P, \theta, \mathcal{H}_{k-1}) &= \int P(\pi_k | \mathcal{H}_{k-1}) \cdot P(\tau_k | \pi_k, P, \theta) d\pi_k \\ &= \mathbb{E}_{\pi_k | \mathcal{H}_{k-1}} \left[\prod_{h=1}^H \pi_k(a_h^k | s_h^k) \right] \cdot L_P(\tau_k; P) \cdot \prod_{i \in [n]} L_{\Theta, i}(\tau_k; \theta_i). \end{aligned}$$

The factors L_P and $L_{\Theta, i}$ depend respectively on P alone and θ_i alone, and the leading expectation depends on τ_k and $\mathcal{H}_{k-1}$ alone. $\square$

A.2.2 Main Theorem

Theorem A.8 (Posterior Decoupling). *Under Assumptions A.1, A.2, and A.3, the posterior distribution over the latent world factorizes at every episode:*

$$P(P, \theta | \mathcal{H}_k) = \mu_{k+1}^P(P) \cdot \prod_{i \in [n]} \mu_{k+1}^{\Theta, i}(\theta_i), \quad \forall k \in \{0, 1, \dots, K\} \quad (12)$$

where the marginals admit the recursive updates

$$\mu_{k+1}^P(P) \propto \mu_k^P(P) \cdot L_P(\tau_k; P), \quad \mu_{k+1}^{\Theta,i}(\theta_i) \propto \mu_k^{\Theta,i}(\theta_i) \cdot L_{\Theta,i}(\tau_k; \theta_i) \quad (13)$$

with initial conditions $\mu_1^P = P_0^P$ and $\mu_1^{\Theta,i} = P_0^{\Theta,i}$.

Proof. We proceed by induction on k .

Base case ($k = 0$). By Assumption A.3, $P(P, \theta | \mathcal{H}_0) = P_0(P, \theta) = P_0^P(P) \prod_{i \in [n]} P_0^{\Theta,i}(\theta_i)$, which has the required factored form.

Inductive step. Suppose $P(P, \theta | \mathcal{H}_{k-1}) = \mu_k^P(P) \prod_i \mu_k^{\Theta,i}(\theta_i)$. By Bayes' rule,

$$P(P, \theta | \mathcal{H}_k) \propto P(P, \theta | \mathcal{H}_{k-1}) \cdot P(\tau_k | P, \theta, \mathcal{H}_{k-1}). \quad (14)$$

Apply Lemma A.7:

$$\begin{aligned} P(P, \theta | \mathcal{H}_k) &\propto \mu_k^P(P) \prod_i \mu_k^{\Theta,i}(\theta_i) \cdot c_k(\tau_k) \cdot L_P(\tau_k; P) \cdot \prod_i L_{\Theta,i}(\tau_k; \theta_i) \\ &\propto [\mu_k^P(P) L_P(\tau_k; P)] \cdot \prod_i [\mu_k^{\Theta,i}(\theta_i) L_{\Theta,i}(\tau_k; \theta_i)]. \end{aligned}$$

The factor $c_k(\tau_k)$ does not depend on (P, θ) and is absorbed into the normalization constant. The remaining product factorizes into a P -only term and n separate θ_i -only terms. Normalizing each factor independently yields the recursive forms stated in the theorem and establishes the desired posterior factorization. $\square$

Corollary A.9 (Storage and Update Complexity). *Maintaining the joint posterior $P(P, \theta | \mathcal{H}_k)$ under PACT requires $O(\dim(P_0^P) + n|\Theta|)$ storage, as opposed to the naive $O(\dim(P_0^P) \cdot |\Theta|^n)$ that a non-factored posterior would require. Each Bayesian update costs $O(H)$ likelihood evaluations per module.*

Proposition A.10 (Trajectory-Level Equivalence with Joint-PSRL). *Consider an idealised Joint-PSRL algorithm that maintains the joint posterior μ_k^Θ on $\Theta = \prod_i \Theta_i$ and at each episode samples $\theta_k \sim \mu_k^\Theta$. Under the assumptions of Theorem A.8 and coupled RNG (formalised below), the trajectories produced by PACT and Joint-PSRL are path-wise identical: $\tau_k^{\text{H-PSMG}} = \tau_k^{\text{Joint-PSRL}}$ for every $k \geq 1$ almost surely. In particular the cumulative regrets agree exactly:*

$$\text{Reg}_i^B[\text{PACT}](K) = \text{Reg}_i^B[\text{Joint-PSRL}](K) \quad \forall i, K, \text{ a.s.} \quad (15)$$

The coupling is constructed as follows. At episode k , both algorithms draw a single shared seed $u_k = (u_k^P, u_k^{\Theta,1}, \dots, u_k^{\Theta,n}) \sim \text{Unif}([0, 1]^{n+1})$. PACT uses u_k^P to sample $\hat{P}_k = F_{\mu_k^P}^{-1}(u_k^P)$ and $u_k^{\Theta,i}$ to sample $\hat{\theta}_i^k = F_{\mu_k^{\Theta,i}}^{-1}(u_k^{\Theta,i})$ via the inverse-CDF transform. Joint-PSRL uses u_k^P identically for $\hat{P}_k$ and the joint draw $\hat{\theta}_k = F_{\mu_k^\Theta}^{-1}(u_k^{\Theta,1}, \dots, u_k^{\Theta,n})$ via the per-coordinate inverse-CDF of the product factorisation.

Proof. By Theorem A.8, $\mu_k^\Theta(\theta_1, \dots, \theta_n) = \prod_i \mu_k^{\Theta,i}(\theta_i)$ at every k . The inverse-CDF of a product distribution applied coordinate-wise yields independent samples from the marginals. Hence the two algorithms produce identical $\hat{P}_k$ and identical $(\hat{\theta}_1^k, \dots, \hat{\theta}_n^k)$, hence identical CCE policies $\pi_k = \text{CCE}(\hat{P}_k, \hat{\theta}_k)$ (Assumption A.6: deterministic oracle), hence identical trajectories τ_k under shared environment randomness. Identical trajectories imply identical posteriors at $k+1$ by Bayes' rule, and the argument repeats inductively. $\square$

Remark A.11. *The decoupling is more than a notational convenience. It separates two distinct sources of uncertainty into modular inference problems. Model uncertainty is captured by μ^P , which behaves exactly as in single-agent PSRL and inherits all known concentration tools (Agrawal and Jia, 2017). Type uncertainty is captured by $\mu^{\Theta,i}$, which is a categorical Bayesian estimation problem amenable to information-theoretic analysis (Russo and Van Roy, 2014). This separation is the basis of the regret decomposition outlined in Section A.5.*

A.3 Numerical Illustration: Two-Agent Bayesian Prisoner's Dilemma

To make the decoupling concrete we work through a minimal instance numerically. The example is a stateless two-agent two-type matrix game played for K rounds.

A.3.1 Setup

Set $|\mathcal{S}| = 1$, $\mathcal{A}_1 = \mathcal{A}_2 = \{C, D\}$, and $\Theta_i = \{P, S\}$ for $i \in \{1, 2\}$, where P denotes pro-social and S denotes selfish. The reward of agent i as a function of its type is

$$r_i^S(a_i, a_{-i}) = \mathbb{1}[a_i = D], \quad r_i^P(a_i, a_{-i}) = \mathbb{1}[a_i = C \wedge a_{-i} = C]. \quad (16)$$

A selfish agent prefers defection. A pro-social agent rewards mutual cooperation. The horizon is $H = 1$ and there is no transition uncertainty, so the model component μ^P is trivial and the example isolates the type-inference component.

The prior is uniform: $\mu_1^{\Theta,1}(P) = \mu_1^{\Theta,1}(S) = 0.5$ and similarly for agent 2, so $\mu_1^\Theta(\theta) = 0.25$ for each $\theta \in \Theta$. The true type profile is $\theta^* = (P, S)$.

A.3.2 Episode 1

The algorithm draws $\hat{\theta}_1 \sim \mu_1^\Theta$ uniformly. Suppose the realization is $\hat{\theta}_1 = (P, P)$. The CCE of the all-pro-social game is the pure strategy (C, C) . The algorithm executes (C, C) , and rewards are realized in the true world: $r_1 = r_1^P(C, C) = 1$ and $r_2 = r_2^S(C, \cdot) = 0$.

The reward likelihoods are

$$L_{\Theta,1}(\tau_1; \theta_1) = P(r_1 = 1 | a = (C, C), \theta_1) = \begin{cases} 1 & \theta_1 = P \\ 0 & \theta_1 = S \end{cases} \quad (17)$$

$$L_{\Theta,2}(\tau_1; \theta_2) = P(r_2 = 0 \mid a = (C, C), \theta_2) = \begin{cases} 0 & \theta_2 = P \\ 1 & \theta_2 = S \end{cases}. \quad (18)$$

Table 3 tabulates the posterior update.

Table 3: Posterior update after episode 1. Joint posterior is computed by Bayes’ rule, then verified against the product of marginals.

θ_1	θ_2	$\mu_1^{\Theta}(\theta)$	$L_{\Theta,1}$	$L_{\Theta,2}$	Unnormalized posterior
P	P	0.25	1	0	0
P	S	0.25	1	1	0.25
S	P	0.25	0	0	0
S	S	0.25	0	1	0
Normalized					
P	S				1.00
others					0.00

The joint posterior puts probability 1 on the true type profile (P, S) . We verify the decoupling by computing the marginals:

$$\mu_2^{\Theta,1}(P) = \mu_1^{\Theta,1}(P) \cdot L_{\Theta,1}(\tau_1; P) / Z_1 = 0.5 \cdot 1 / 0.5 = 1, \quad (19)$$

$$\mu_2^{\Theta,1}(S) = 0, \quad (20)$$

$$\mu_2^{\Theta,2}(P) = \mu_1^{\Theta,2}(P) \cdot L_{\Theta,2}(\tau_1; P) / Z_2 = 0.5 \cdot 0 / 0.5 = 0, \quad (21)$$

$$\mu_2^{\Theta,2}(S) = 1. \quad (22)$$

The product $\mu_2^{\Theta,1} \cdot \mu_2^{\Theta,2}$ equals the unnormalized joint $\delta_{(P,S)}$, confirming the decoupling in Theorem A.8.

A.3.3 Noisy Observation Variant

To show that decoupling is not an artifact of deterministic rewards, suppose now the reward signal flips with probability $\epsilon = 0.1$. The likelihoods become

$$L_{\Theta,1}(\tau_1; \theta_1) = \begin{cases} 0.9 & \theta_1 = P \\ 0.1 & \theta_1 = S \end{cases}, \quad L_{\Theta,2}(\tau_1; \theta_2) = \begin{cases} 0.1 & \theta_2 = P \\ 0.9 & \theta_2 = S \end{cases}. \quad (23)$$

The unnormalized joint posterior is the outer product $\mu_1^{\Theta} \cdot L_{\Theta,1} \cdot L_{\Theta,2}$ which still factorizes as $\mu_2^{\Theta,1} \cdot \mu_2^{\Theta,2}$ with $\mu_2^{\Theta,1}(P) = 0.9, \mu_2^{\Theta,1}(S) = 0.1$ and $\mu_2^{\Theta,2}(S) = 0.9, \mu_2^{\Theta,2}(P) = 0.1$. Concentration is partial after one observation but the factorization is exact.

A.4 Necessity of the Structural Assumptions

We show that both Assumptions A.1 and A.2 are necessary in the sense that violation of either causes the decoupling to break.

A.4.1 Counter-Example I: Type-Dependent Transitions

Suppose Assumption A.1 fails. Concretely, consider a two-state HT-MG where $P(s' \mid s, a, \theta)$ varies with

θ . The trajectory likelihood becomes

$$P(\tau_k \mid P, \theta, \mathcal{H}_{k-1}) = c_k(\tau_k) \cdot \prod_{h=1}^H P(s_{h+1}^k \mid s_h^k, a_h^k, \theta) \cdot \prod_{h,i} P(r_{h,i}^k \mid s_h^k, a_h^k, \theta_i). \quad (24)$$

The transition factor is now jointly a function of P and θ and does not split. Posterior $P(P, \theta \mid \mathcal{H}_k)$ is no longer factorable into a P -marginal and a θ -marginal. Empirically, observing a transition $(s, a) \rightarrow s'$ simultaneously updates beliefs about both the kernel and the types, with no way to attribute the change to one source alone.

A.4.2 Counter-Example II: Non-Local Reward

Suppose Assumption A.2 fails. Consider a social-comparison reward $r_1(s, a, \theta) = r_1^{\theta_1}(s, a) - \alpha \cdot r_2^{\theta_2}(s, a)$ for some $\alpha > 0$, so that agent 1’s utility includes an envy term proportional to agent 2’s realized return under its own type. Then

$$L_{\Theta,1}(\tau_k; \theta) = \prod_{h=1}^H P\left(r_h^{k,1} - \alpha r_h^{k,2,\theta_2} \mid s_h^k, a_h^k, \theta_1\right) \quad (25)$$

contains θ_2 inside the agent-1 likelihood factor. The likelihood factorization in Lemma A.7 fails. The posterior couples θ_1 and θ_2 at every update step.

These two counter-examples confirm that the decoupling theorem is tight in its assumptions and provide guidance on when the centralized PACT framework is appropriate.

A.5 From Decoupling to Regret: A Roadmap

The decoupling theorem is the structural foundation for a regret bound on PACT. We outline the derivation strategy in this section and provide the rigorous proof in Appendix A.8.

Roadmap. The full proof of the regret bound is given in Appendix A.8, following the per-episode decomposition $\Delta_i^k = \Delta_i^{k,R} + \Delta_i^{k,P}$ established by the value-difference lemma for HT-MG (Lemma A.21): a transition-misspecification term controlled via Hölder + Dirichlet ℓ_1 concentration (Proposition A.29) and a reward-misspecification (type-inference) term controlled via the Russo–Van Roy information-ratio framework on the per-agent meta-bandit on Θ_i (Proposition A.23). We state here the resulting bound and its gap-dependent corollary, then turn to the algorithmic consequences (rate separation, PACT⁺, decoupling lower bound, minimax matching).

A.5.1 Main Regret Bound

Combining the two components and the CCE oracle approximation gives the formal regret bound, which we state here and prove rigorously in Appendix A.8.

Theorem A.12 (Bayesian Regret of PACT, informal). *Under Assumptions A.1, A.2, A.3, A.5, A.6,*

a Dirichlet prior on P^* , and a bounded type information ratio $\bar{\Gamma} = O(H^2|\Theta_i|)$, the Bayesian regret of agent i over K episodes satisfies

$$\mathbb{E}[\text{Reg}_i^B(K)] \leq \tilde{O}(H^{3/2}\sqrt{S \cdot A_{\text{joint}} \cdot K}) + O(H\sqrt{|\Theta_i| \log |\Theta_i| \cdot K}) + K \cdot \epsilon_{\text{CCE}}, \quad (26)$$

where $A_{\text{joint}} = \prod_i |\mathcal{A}_i|$ is the joint action space size and ϵ_{CCE} is the CCE oracle approximation error.

The bound separates the cost of learning the transition kernel from the cost of identifying the type profile. The state-space size S enters only the model-learning component, reflecting the agent-level decoupling of Theorem A.8: each agent's type inference depends only on its own type space Θ_i rather than on the full joint type space Θ . Appendix A.8 provides the full proof, including the value-difference lemma for HT-MG, the information-ratio analysis of type inference, and the Dirichlet concentration argument for the transition component.

Corollary A.13 (Gap-Dependent Rate under Identifiability). *If in addition Assumption A.4 holds with parameter $\rho > 0$, then the type-inference component collapses to a constant in K :*

$$\sum_{k=1}^K \mathbb{E}[\Delta_i^{k,R}] \leq O\left(\frac{H \log |\Theta_i|}{\rho}\right). \quad (27)$$

Consequently, the Bayesian regret simplifies to

$$\mathbb{E}[\text{Reg}_i^B(K)] \leq \tilde{O}(H^{3/2}\sqrt{SA_{\text{joint}}K}) + O\left(\frac{H \log |\Theta_i|}{\rho}\right) + K \cdot \epsilon_{\text{CCE}}. \quad (28)$$

In the asymptotic regime $K \rightarrow \infty$ with $\epsilon_{\text{CCE}} = O(K^{-1/2})$, the bound is dominated by transition learning at rate $\tilde{O}(\sqrt{K})$. The type-inference cost is paid only during a short burn-in period of length $O(\log |\Theta_i|/\rho)$.

The corollary captures an important practical regime: when types are sufficiently distinct (TID with ρ bounded away from zero), the cost of heterogeneity is essentially a one-time identification cost rather than a recurring exploration penalty. Proof appears in Appendix A.8, immediately following Proposition A.23.

A.5.2 Rate Separation from Heuristic and Type-Agnostic Baselines

The $\tilde{O}(\sqrt{K})$ rate of Theorem A.12 is not just a constant-factor improvement over heuristic baselines: the gap is in the asymptotic rate. We make this precise.

Theorem A.14 (Rate Separation). *There exists an HT-MG instance $\mathcal{G}^*$ satisfying Assumptions A.1, A.2, A.3, A.5, A.6, and a uniform prior over types $\mu_0^{\Theta,i} = \text{Unif}(\Theta_i)$, on which the cumulative Bayesian regret of three algorithms satisfies:*

- (i) **PACT (Algorithm 1)**: $\mathbb{E}[\text{Reg}_i^B(K)] = \tilde{O}(\sqrt{K})$ for all $K \geq 1$.

- (ii) **MAP-Type-Greedy** (substitute the MAP estimate $\hat{\theta}_i^k = \arg \max_{\theta} \mu_k^{\Theta,i}(\theta)$ for the posterior sample in Algorithm 1): on a positive-probability event $\mathcal{E}_{\text{trap}}$ (over initial randomness), $\mathbb{E}[\text{Reg}_i^B(K) \mid \mathcal{E}_{\text{trap}}] = \Omega(K)$. In particular, $\mathbb{E}[\text{Reg}_i^B(K)] = \Omega(K)$.

- (iii) **PSRL-NoType** (sample $\hat{\theta}_k \sim \text{Unif}(\Theta)$ every episode, never update the type posterior): $\mathbb{E}[\text{Reg}_i^B(K)] = \Omega(K)$.

Proof Sketch. We construct $\mathcal{G}^*$ explicitly as the exploration-trap game: 5 states $\{0, 1, 2, 3, 4\}$, $n = 2$ agents with action set $\{L, R\}$, types $\Theta_i = \{A, B\}$, horizon H , deterministic majority-vote transitions $((R, R) \mapsto s+1, (L, L) \mapsto s-1, \text{mixed} \mapsto s, \text{clipped at boundaries})$, initial state $s_1 = 0$, and rewards

$$r_i(s, a, A) = 0.5 \quad \text{for all } s, \quad (29)$$

$$r_i(s, a, B) = \begin{cases} 0.5 & \text{if } s \in \{0, 1, 2, 3\}, \\ 1.0 & \text{if } s = 4, \end{cases} \quad (30)$$

with small Gaussian observation noise $\sigma \rightarrow 0$. The true types are $\theta^* = (B, B)$.

(i) **PACT**. This is the direct application of Theorem A.12 to $\mathcal{G}^*$.

(ii) **MAP-Type-Greedy**. Let $\mathcal{E}_{\text{trap}}$ be the event that the initial MAP estimate of agent 1 is A . Under a uniform prior with deterministic tie-breaking (e.g., $\arg \max$ returning the lowest-index maximizer), this event holds with probability 1; with random uniform tie-breaking it has probability $\geq (1/|\Theta_1|)^n = 1/4$.

On $\mathcal{E}_{\text{trap}}$, by induction on k : (1) $\hat{\theta}_{\text{MAP}}^k = (A, A)$; (2) under type A , $V_h^\pi(s; A) = 0.5(H-h+1)$ is independent of state and policy (uniform reward), so all joint actions yield identical Q -values and the CCE-with-Nash-selection picks $\pi(L, L)$ at every (s, h) by tie-breaking; (3) rollout from $s_1 = 0$ under deterministic transitions stays at $s = 0$ for all H steps; (4) at state 0 both candidate types predict $r_h^i = 0.5$, so the log-likelihood ratio $\log \mu_{k+1}^{\Theta,i}(A)/\mu_{k+1}^{\Theta,i}(B) = \log \mu_k^{\Theta,i}(A)/\mu_k^{\Theta,i}(B) + O(\sigma)$, which converges in the limit $\sigma \rightarrow 0$; (5) the MAP estimate remains A .

The per-episode value of MAP-Type-Greedy on $\mathcal{E}_{\text{trap}}$ is $V^{\text{MAP}} = 0.5H$ per agent. The oracle value satisfies $V^* \geq H - 4 + 1 \cdot (H - 4) = H - 4 + (H - 4)$ approximately (the optimal (B, B) policy reaches state 4 in 4 steps and earns $r = 1$ for the remaining $H - 4$ steps, with $r = 0.5$ during the traverse), giving $V^* \geq 0.5 \cdot 4 + 1.0 \cdot (H - 4) = H - 2$ per agent for $H \geq 4$. The per-episode regret is $V^* - V^{\text{MAP}} \geq H - 2 - 0.5H = 0.5H - 2 > 0$ for $H > 4$. Summed over K episodes, $\mathbb{E}[\text{Reg}_i^B(K) \mid \mathcal{E}_{\text{trap}}] \geq (0.5H - 2)K = \Omega(K)$, completing the claim.

(iii) **PSRL-NoType**. The algorithm samples $\hat{\theta}_k \sim \text{Unif}(\Theta_1 \times \Theta_2)$ every episode, independent of history. Let $V(\theta) := \mathbb{E}^{\pi(\theta), P^*}[\sum_h r_i(s_h, a_h, \theta_i^*)]$ be

the value of the CCE for sampled types θ evaluated under the true types $\theta^* = (B, B)$. Then the expected per-episode value is $\bar{V} = \mathbb{E}_{\theta \sim \text{Unif}}[V(\theta)]$. By a direct case analysis, $\bar{V} < V^*$ for the trap game, since the three non- (B, B) joint types ($\frac{3}{4}$ of samples) induce policies that fail to reach state 4. The per-episode regret is the positive constant $V^* - \bar{V}$, giving $\mathbb{E}[\text{Reg}_i^B(K)] = (V^* - \bar{V})K = \Omega(K)$. $\square$

The theorem makes precise the qualitative claim that Bayesian posterior sampling on types provides exploration that heuristic alternatives lack. The proof construction is the exploration-trap game $\mathcal{G}^*$ used to drive the deep-trap family $\mathcal{G}_m^{\text{trap}}$ analysed in Section A.5.3; the empirical counterpart of this separation, in the LLM-tier setting where action spaces and reward observations come from prompted backbones rather than analytic kernels, is reported in Section 4.

A.5.3 Regret guarantees of PACT⁺

The algorithm PACT⁺ (Algorithm 2) was introduced in Section 3. This subsection establishes its two regret guarantees: asymptotic match with PACT (worst-case rate unchanged) and exponential-in- $|\Theta|^n$ burn-in improvement on the deep-trap family. Both are direct corollaries of the type-discrimination bonus structure and the posterior decoupling theorem.

Theorem A.15 (Regret of PACT⁺). *Let $\beta > 0$. Under the assumptions of Theorem A.12:*

(i) **Asymptotic match.** *On any HT-MG, the Bayesian regret of PACT⁺ satisfies*

$$\mathbb{E}[\text{Reg}_i^B(K)] \leq \tilde{O}(H^{3/2} \sqrt{SA_{\text{joint}}K} + H\sqrt{|\Theta_i|K}), \quad (31)$$

identical to the PACT bound of Theorem A.12.

(ii) **Exponential-in- $|\Theta|^n$ burn-in improvement.** *Let $\mathcal{G}_m^{\text{trap}}$ denote the family of deep-trap HT-MGs parameterised by per-agent type-space size $|\Theta_i| = m$ (Appendix B for the precise definition: n agents, $S = m + 2$ states, deterministic unanimity-vote transitions, horizon H , single jackpot type θ_{jack} with $r_i(s^*, a, \theta_{\text{jack}}) = 1$ at the rightmost state and $r_i = 0.5$ otherwise, uniform prior, true types $\theta_i^* = \theta_{\text{jack}}$). Under Assumption B.1 (welfare-maximising CCE oracle, lowest-index tie-break), the expected burn-in regrets are:*

$$\begin{aligned} \text{PACT} / \text{Joint-PSRL: } & \mathbb{E}[\text{Reg}_i^B(\tau_{\text{PACT}})] = \Theta(m^n \cdot H), \\ \text{PACT}^+ \text{ (any } \beta > 0): & \mathbb{E}[\text{Reg}_i^B(\tau_{\text{PACT}^+})] = O(H). \end{aligned}$$

The burn-in separation is $\Omega(m^n)$, exponential in the number of agents.

Proof Sketch. (i) **Asymptotic match.** By a CCE-optimality decomposition (Lemma B.8, App B.0.8), the per-episode regret of PACT⁺ relative to PACT is bounded by the expected cumulative bonus collected on the rollout, $\beta H \cdot \mathbb{E}[\max_{s,a} D_k(s, a)]$. A

collision-probability argument (Lemma B.9) gives $D_k(s, a) \leq 2 \sum_i (1 - \mu_k^{\Theta_i, i}(\theta_i^*))$, reducing the cumulative bonus to a sum of posterior-error probabilities. Under Assumption A.4 (Hellinger gap $\rho > 0$), standard Bayesian posterior concentration on finite type spaces (Ghosal and van der Vaart, 2017) gives an exponential decay $\mathbb{E}[1 - \mu_k^{\Theta_i, i}(\theta_i^*)] \leq |\Theta_i| \exp(-\rho H k / 2)$, so the cumulative bonus regret is bounded by a K -independent constant $O(\beta n |\Theta_i| / \rho)$ (Proposition B.10). This is absorbed into the leading $\tilde{O}(H \sqrt{|\Theta_i|K})$ type-inference term of Theorem A.12.

(ii) **Exponential burn-in.** For PACT: under uniform prior, $\Pr[\hat{\theta}_k = (\theta_{\text{jack}})^n] = m^{-n}$ per episode, independent across k . Before any successful draw, every visited state $s \in \{0, 1, \dots, S-2\}$ has identical predicted reward 0.5 across all types, so observations carry no information about types (Lemma B.5); the marginal stays uniform. The expected wait for the first informative draw is $\text{Geom}(m^{-n})$ with mean m^n ; each stuck episode contributes per-episode regret $\Theta(H)$, totalling $\Theta(m^n H)$ (Proposition B.6).

For PACT⁺: at uniform prior, the bonus evaluates to $D_k(s^*, a) = n(m-1)/m^2 > 0$ at the jackpot state and 0 elsewhere (Equation 56). Backward induction with this bonus and deterministic unanimity transitions gives social-welfare gap

$$W(a^R) - W(a^L) = n(H - S + 1)(0.5\alpha + \beta n p_k(1 - p_k)) > 0$$

regardless of the sampled $\hat{\theta}$ (Lemma B.3), where α is the fraction of jackpot-sampled agents. Under Assumption B.1, the CCE oracle returns a^R ; the trajectory reaches s^* deterministically; the marginal concentrates on θ_{jack} within one episode (Proposition B.7). Burn-in regret is $O(H)$, independent of m and n .

The full formal proof is in Appendix B. $\square$

Comparison with Joint-PSRL. Theorem A.15 formalises the algorithmic gap between PACT⁺ and Joint-PSRL. Although vanilla PACT and Joint-PSRL produce bit-identical trajectories under matched RNG (a consequence of Theorem A.8), the bonus D_k requires per-agent factored marginals, which Joint-PSRL only computes by marginalising the joint posterior at $O(|\Theta|^n)$ cost per query. The decoupling theorem therefore enables three distinct improvements over Joint-PSRL:

- **Storage.** $O(n|\Theta_i|)$ instead of $\Omega(|\Theta|^n)$ posterior memory (Corollary A.9). For $|\Theta_i| = 4$, the joint posterior grows from 16 entries at $n = 2$ to 1024 entries at $n = 5$, a $64\times$ gap that we measure empirically on an LLM substrate in Section 4.
- **Regret analysis.** $\tilde{O}(\sqrt{|\Theta_i|K})$ type-inference rate instead of $\tilde{O}(\sqrt{|\Theta|^n K})$, the generic

Bayesian-RL guarantee (Theorem A.16, Section A.5.4).

- **Algorithmic refinement.** Exponential-in- $|\Theta|^n$ burn-in regret improvement of PACT⁺ over Joint-PSRL on the deep-trap family (Theorem A.15(ii)).

The empirical separation, with PACT⁺’s burn-in benefit growing monotonically in $|\Theta_i|$ and n on an LLM substrate, is reported in Section 4.

A.5.4 Exponential improvement over generic Bayesian RL via decoupling

Theorem A.12 produces a regret bound whose type-inference cost depends on $|\Theta_i|$, the per-agent type-space size, rather than on $|\Theta| = \prod_i |\Theta_i|$, the joint type-space size. A natural question is whether this dependence is intrinsic to the HT-MG problem or an artefact of our structural assumptions. We answer it by showing that without Assumption A.3, no algorithm can match the bound: the best achievable Bayesian regret depends on $|\Theta|^n$, exponentially worse than ours.

Theorem A.16 (Exponential Lower Bound Without Prior Factorisation). *Let $\mathcal{F}_{n,m,S,A_{\text{joint}},H}^{\text{noPF}}$ denote the family of HT-MG instances with n agents, $|\Theta_i| = m$ for each i , state-space size S , joint-action-space size A_{joint} , horizon H , satisfying Assumptions A.1, A.2, A.5, A.6, but for which the joint type prior μ_0^Θ is permitted to be an arbitrary distribution over $\Theta = \prod_i \Theta_i$ (i.e., A.3 is dropped). For any algorithm $\mathfrak{A}$,*

$$\sup_{\mathcal{G} \in \mathcal{F}^{\text{noPF}}} \mathbb{E}_{\mathcal{G}, \mathfrak{A}} [\text{Reg}_i^B(K)] = \Omega(\sqrt{m^n \cdot K}). \quad (32)$$

Proof Sketch. We reduce a Bayesian multi-armed bandit with $M = m^n$ arms to an HT-MG without prior factorisation. Take $S = 1$, $H = 1$, $A_{\text{joint}} = M$. For each joint type $\theta \in \Theta$, choose a one-to-one map $a^*: \Theta \rightarrow \mathcal{A}_{\text{joint}}$ and define the reward as $r(a, \theta) = \mathbb{1}[a = a^*(\theta)]$. Place the prior μ_0^Θ uniform on Θ but not factored across agents (e.g., concentrated on a uniformly random permutation of Θ). Conditional on θ the unique optimal joint action is $a^*(\theta)$; without prior factorisation, observing one agent’s marginal reward provides no information about another agent’s type that is not already captured by the joint posterior over Θ .

The resulting problem is a Bayesian M -armed bandit with uniform prior over the identity of the optimal arm. By the standard Bayesian regret lower bound for multi-armed bandits (Lattimore and Szepesvári, 2020, Theorem 15.2), $\mathbb{E}_{\mathfrak{A}} [\text{Reg}(K)] = \Omega(\sqrt{MK}) = \Omega(\sqrt{m^n K})$. Per-agent regret equals total regret in this single-agent-reduces-to-bandit construction up to a factor of n , completing the claim. $\square$

Theorem A.16 combined with Theorem A.12 establishes an exponential separation. With prior factorisation (Assumption A.3), PACT achieves type-inference regret $\tilde{O}(H\sqrt{mK})$; without it, no algorithm achieves better than $\Omega(\sqrt{m^n K})$. The improvement factor in the bound is $m^{(n-1)/2}$, exponential in the number of agents. The structural assumption set TI + RL + PF is therefore not merely a convenience for analysis but a necessary condition for sublinear-in- n dependence on the joint type space.

Remark A.17 (Comparison with Joint-PSRL). *A particularly useful consequence is the regret bound for Joint-PSRL implied by our analysis. The naive Bayesian-RL analysis of Joint-PSRL on the joint hypothesis class Θ would invoke the generic information-ratio bound for posterior sampling on a hypothesis class of size $|\Theta|^n$, yielding a regret bound of $\tilde{O}(H\sqrt{|\Theta|^n K})$. Our Theorem A.8 (Decoupling) shows that under TI + RL + PF the joint posterior is the product of marginals at every episode, which when combined with the per-agent information ratio of Lemma A.24 reduces the Joint-PSRL regret bound to the same $\tilde{O}(H\sqrt{|\Theta_i| K})$ as PACT (Theorem A.12). Thus our decoupling theorem provides an exponential-in- n sharpening of the Joint-PSRL regret guarantee at no cost in storage or computation for Joint-PSRL (which is already storing the joint posterior); for PACT, the additional benefit is the exponential storage reduction documented analytically by Corollary A.9 and verified empirically on an LLM substrate in Section 4.*

A.5.5 Minimax lower bound and tightness of Theorem A.12

Within the family of HT-MGs satisfying all our assumptions, we establish a minimax lower bound that nearly matches Theorem A.12.

Theorem A.18 (Minimax Bayesian Regret Lower Bound under PF). *Let $\mathcal{F}_{n,m,S,A_{\text{joint}},H}^{\text{PF}}$ denote the family of HT-MG instances satisfying Assumptions A.1, A.2, A.3, A.5, A.6. For any algorithm $\mathfrak{A}$ and parameters $K \geq SA_{\text{joint}}H$ and $K \geq mH$, there exists an instance $\mathcal{G} \in \mathcal{F}^{\text{PF}}$ with*

$$\mathbb{E}_{\mathcal{G}, \mathfrak{A}} [\text{Reg}_i^B(K)] = \Omega\left(\sqrt{H^3 SA_{\text{joint}} K} + \sqrt{|\Theta_i| HK}\right). \quad (33)$$

Proof Sketch. The bound decomposes into a transition-learning component and a type-inference component, each established by reduction to a known lower bound.

Transition component $\Omega(\sqrt{H^3 SA_{\text{joint}} K})$. Specialise to $|\Theta_i| = 1$, eliminating type uncertainty. The HT-MG reduces to a standard finite-horizon MDP with state space $\mathcal{S}$, joint action space $\mathcal{A}_{\text{joint}}$, horizon H . The minimax lower bound for episodic RL of Domingues et al. (2021, Theorem 1) gives expected cumulative regret $\Omega(\sqrt{H^3 SAK})$ for any

algorithm; substituting $A = A_{\text{joint}}$ completes this component.

Type-inference component $\Omega(\sqrt{|\Theta_i|HK})$. Specialise to $S = 1$, $A_{\text{joint}} = |\Theta_i|$, $|\Theta_j| = 1$ for $j \neq i$ (no uncertainty about other agents' types). Let $r_i(s, a, \theta_i) = \mathbb{1}[a = a^*(\theta_i)]$ for a one-to-one mapping $a^*: \Theta_i \rightarrow \mathcal{A}_{\text{joint}}$. The resulting problem is a stochastic multi-armed bandit with $|\Theta_i|$ arms and $T = KH$ rounds (each episode provides H reward observations of the same type). The standard MAB minimax lower bound (Lattimore and Szepesvári, 2020, Theorem 15.2) gives $\Omega(\sqrt{|\Theta_i|T}) = \Omega(\sqrt{|\Theta_i|HK})$.

The two constructions can be combined into a single HT-MG with both transition and type uncertainty; the resulting lower bound is the maximum of the two components, which equals their sum up to a constant factor. $\square$

Comparing the lower bound of Theorem A.18 with the upper bound of Theorem A.12:

Component	Upper bound (Thm. A.12)	Lower bound (Thm. A.18)
Transition	$\tilde{O}(\sqrt{H^3 S A_{\text{joint}} K})$	$\Omega(\sqrt{H^3 S A_{\text{joint}} K})$
Type inference	$\tilde{O}(\sqrt{H^2 \Theta_i K})$	$\Omega(\sqrt{H \Theta_i K})$

The transition term is tight modulo logarithmic factors. The type-inference term has a $\sqrt{H}$ gap, consistent with similar gaps in Bayesian-RL upper bounds and conjectured (but unresolved) to be closable via variance-aware Bernstein-type analyses. Theorem A.16 establishes that without PF this entire analysis collapses: the lower bound jumps from $\sqrt{|\Theta_i|}$ to $\sqrt{|\Theta|^n}$, an exponential-in- n inflation. Our regret bound is thus near-optimal among algorithms that exploit the structural assumptions of HT-MG.

A.6 Verification of $c_k(\tau_k)$ Independence of (P, θ)

We elaborate on the claim in Lemma A.7 that $c_k(\tau_k)$ does not depend on (P, θ) .

The policy π_k is a random variable taking values in $\Delta(\mathcal{A})^S$. Its distribution $P(\pi_k \mid \mathcal{H}_{k-1})$ is induced by the algorithm:

$$P(\pi_k \mid \mathcal{H}_{k-1}) = \int \mathbb{1}[\pi_k = \text{CCE}(\hat{P}, \hat{\theta})] d\mu_k^P(\hat{P}) \prod_i d\mu_k^{\Theta_i}(\hat{\theta}_i). \quad (34)$$

Both μ_k^P and $\mu_k^{\Theta_i}$ are measurable with respect to $\mathcal{H}_{k-1}$. By the tower property of conditional expectation, $P(\pi_k \mid \mathcal{H}_{k-1}, P, \theta) = P(\pi_k \mid \mathcal{H}_{k-1})$, since the algorithmic randomness in choosing π_k is conditionally independent of the true latent given the history. Hence

$$c_k(\tau_k) = \int P(\pi_k \mid \mathcal{H}_{k-1}) \prod_{h=1}^H \pi_k(a_h^k \mid s_h^k) d\pi_k \quad (35)$$

is a function of τ_k and $\mathcal{H}_{k-1}$ alone.

This argument reflects a standard property of posterior sampling: the algorithm uses only the posterior, not the truth, to choose its policy. The truth enters the system only through the realized trajectory.

A.7 Discussion of CCE Oracle Approximation

In Algorithm 1, the CCE oracle is assumed to return an ϵ_{CCE} -approximate CCE. We comment on how this approximation enters the regret bound.

Formal oracle versus reported implementations. The formal oracle returns a welfare-maximising CCE in a finite-horizon Markov game. The reported HP-SPGG runner does not solve a CCE linear program: it evaluates all finite joint contribution profiles and applies a deterministic `argmax`; compact Concordia likewise enumerates the finite action menu, using either focal payoff or total payoff as explicitly labelled. These are exact centralised planners for their reported finite objectives, with zero enumeration error, but they should not be read as general CCE implementations. SOTOPIA-Hard has no CCE layer at all because utterances are open-ended LLM generations. The candidate-budget diagnostic in Appendix C.3 measures empirical search error separately from posterior error.

Let π_k be the policy returned by the oracle on the sampled world $(\hat{P}_k, \hat{\theta}_k)$. Then

$$\max_{\pi' \in \text{CCE}(\hat{P}_k, \hat{\theta}_k)} V^{\pi'}(s_1; \hat{P}_k, \hat{\theta}_k) - V^{\pi_k}(s_1; \hat{P}_k, \hat{\theta}_k) \leq \epsilon_{\text{CCE}}. \quad (36)$$

Here ϵ_{CCE} denotes objective suboptimality within the feasible CCE set; equilibrium-feasibility violation should be reported separately by the maximum unilateral-deviation incentive. The objective error contributes additively across K episodes, yielding the $K\epsilon_{\text{CCE}}$ term in (26). Standard V-learning style oracles (Jin et al., 2021) achieve decreasing approximation error as the inner budget grows. LLM latency does not enter this planner term: it enters only through action generation or observation-model calls outside the tabular solve.

A.8 Per-Episode Regret Decomposition: Full Derivation

This appendix provides the rigorous decomposition of per-episode regret into a transition-misspecification component and a reward-misspecification component. The decomposition is the technical bridge between Theorem A.8 (posterior decoupling) and the cumulative regret bound sketched in Section A.5.

A.8.1 Notation and Setup

We use the value function notation $V_{i,h}^\pi(s; P, \theta_i)$ for agent i 's expected return from state s at step h

under policy π , transition kernel P , and own type θ_i . Formally,

$$V_{i,h}^\pi(s; P, \theta_i) := \mathbb{E}_{\pi, P} \left[\sum_{h'=h}^H r_i(s_{h'}, a_{h'}, \theta_i) \mid s_h = s \right], \quad (37)$$

with $V_{i,H+1}^\pi \equiv 0$. The state-action value satisfies the Bellman recursion

$$Q_{i,h}^\pi(s, a; P, \theta_i) = r_i(s, a, \theta_i) + \mathbb{E}_{s' \sim P(\cdot | s, a)} [V_{i,h+1}^\pi(s'; P, \theta_i)]. \quad (38)$$

The episode- k regret of agent i is

$$\Delta_i^k := V_{i,1}^{\pi^*}(s_1^k; P^*, \theta_i^*) - V_{i,1}^{\pi_k}(s_1^k; P^*, \theta_i^*), \quad (39)$$

where $\pi^* := \text{CCE}(P^*, \theta^*)$ is the equilibrium of the true world and $\pi_k := \text{CCE}(\hat{P}_k, \hat{\theta}_k)$ is the equilibrium of the sampled world.

A.8.2 Reduction to the Simulation Gap via Posterior Sampling

The first step is to convert the regret expression into a comparison between the sampled and true value of the executed policy. The reduction relies on a property of posterior sampling formalized by Russo and Van Roy (2014); Osband and Van Roy (2017).

Lemma A.19 (Posterior Sampling Equivalence). *Under Assumptions A.5 and A.6, let f be any measurable function on (P, θ, s_1) , possibly depending on $\mathcal{H}_{k-1}$. Then*

$$\mathbb{E}[f(\hat{P}_k, \hat{\theta}_k, s_1^k) \mid \mathcal{H}_{k-1}] = \mathbb{E}[f(P^*, \theta^*, s_1^k) \mid \mathcal{H}_{k-1}]. \quad (40)$$

Proof. By Assumption A.5, $s_1^k \sim \beta$ is independent of (P^*, θ^*) , so conditioning on s_1^k does not affect the posterior over the latent. By construction of PACT (Algorithm 1), $(\hat{P}_k, \hat{\theta}_k)$ are drawn from the posterior $P(P, \theta \mid \mathcal{H}_{k-1})$, which is, by definition, the conditional distribution of (P^*, θ^*) given $\mathcal{H}_{k-1}$. Hence $(\hat{P}_k, \hat{\theta}_k)$ and (P^*, θ^*) have identical conditional distribution. The equality of expectations follows. Assumption A.6 ensures f is well-defined when it depends on $\text{CCE}(P, \theta)$, since the oracle is a deterministic mapping. $\square$

Lemma A.20 (Reduction to Simulation Gap). *The Bayesian regret of agent i at episode k admits the equality*

$$\mathbb{E}[\Delta_i^k \mid \mathcal{H}_{k-1}] = \mathbb{E}[V_{i,1}^{\pi_k}(s_1^k; \hat{P}_k, \hat{\theta}_i^k) - V_{i,1}^{\pi_k}(s_1^k; P^*, \theta_i^*) \mid \mathcal{H}_{k-1}]. \quad (41)$$

Proof. Apply Lemma A.19 with $f(P, \theta) := V_{i,1}^{\text{CCE}(P, \theta)}(s_1^k; P, \theta_i)$:

$$\mathbb{E}[V_{i,1}^{\pi^*}(s_1^k; P^*, \theta_i^*) \mid \mathcal{H}_{k-1}] = \mathbb{E}[V_{i,1}^{\pi_k}(s_1^k; \hat{P}_k, \hat{\theta}_i^k) \mid \mathcal{H}_{k-1}]. \quad (42)$$

Subtracting $\mathbb{E}[V_{i,1}^{\pi_k}(s_1^k; P^*, \theta_i^*) \mid \mathcal{H}_{k-1}]$ from both sides and rewriting the left side using definition (39) yields (41). $\square$

The reduction shifts the problem from comparing the true optimal value to comparing two evaluations of the same policy, one under the sampled world and one under the true world. This is the standard *simulation gap* in posterior sampling analysis.

A.8.3 Value Difference Lemma for HT-MG

To bound the simulation gap, we adapt the classical simulation lemma (Kearns and Singh, 2002; Bai and Jin, 2020) to HT-MG with the value function indexed by type.

Lemma A.21 (Value Difference for HT-MG). *For any policy π , transition kernels P_1, P_2 , types $\theta_{i,1}, \theta_{i,2}$, and initial state s_1 ,*

$$V_{i,1}^\pi(s_1; P_1, \theta_{i,1}) - V_{i,1}^\pi(s_1; P_2, \theta_{i,2}) = \sum_{h=1}^H \mathbb{E}_{(s,a) \sim d_h^{\pi, P_1}} [\Phi_h^R(s, a) + \Phi_h^P(s, a)], \quad (43)$$

where the per-step reward and transition discrepancies are

$$\begin{aligned} \Phi_h^R(s, a) &:= r_i(s, a, \theta_{i,1}) - r_i(s, a, \theta_{i,2}), \\ \Phi_h^P(s, a) &:= \langle P_1(\cdot \mid s, a) - P_2(\cdot \mid s, a), V_{i,h+1}^\pi(\cdot; P_2, \theta_{i,2}) \rangle, \end{aligned}$$

and d_h^{π, P_1} is the occupancy distribution of (s_h, a_h) under π in P_1 .

Proof sketch. Let $\delta_h(s) := V_{i,h}^\pi(s; P_1, \theta_{i,1}) - V_{i,h}^\pi(s; P_2, \theta_{i,2})$, so $\delta_{H+1} \equiv 0$. Expanding the two Bellman backups and adding/subtracting $\mathbb{E}_{s' \sim P_1}[V_{i,h+1}^\pi(s'; P_2, \theta_{i,2})]$ gives the recursion $\delta_h(s) = \mathbb{E}_{a \sim \pi}[\Phi_h^R + \Phi_h^P + \mathbb{E}_{s' \sim P_1} \delta_{h+1}(s')]$. Unrolling yields (43). The *symmetric form*, used below, swaps the rollout to P_2 and the value weight to $V^\pi(\cdot; P_1, \theta_{i,1})$; this lets us evaluate the inner product along the realized trajectory drawn from P^* rather than under the sampled $\hat{P}_k$ (cf. Proposition A.29). $\square$

Remark A.22 (Symmetric form). *Reversing the add/subtract step gives the equivalent identity with occupancy d_h^{π, P_2} and discrepancy weight $\tilde{\Phi}_h^P(s, a) := \langle P_1 - P_2, V_{i,h+1}^\pi(\cdot; P_1, \theta_{i,1}) \rangle$. This is the form invoked in Step 0 of Proposition A.29.*

Applying Lemma A.21 with $P_1 = \hat{P}_k$, $P_2 = P^*$, $\theta_{i,1} = \hat{\theta}_i^k$, $\theta_{i,2} = \theta_i^*$, and combining with Lemma A.20 yields the central decomposition:

$$\mathbb{E}[\Delta_i^k \mid \mathcal{H}_{k-1}] = \underbrace{\mathbb{E} \left[\sum_{h=1}^H \mathbb{E}_{d_h^{\pi_k, \hat{P}_k}} \Phi_h^R \mid \mathcal{H}_{k-1} \right]}_{\Delta_i^{k,R}} + \underbrace{\mathbb{E} \left[\sum_{h=1}^H \mathbb{E}_{d_h^{\pi_k, P^*}} \Phi_h^P \mid \mathcal{H}_{k-1} \right]}_{\Delta_i^{k,P}}. \quad (44)$$

The reward-misspec term $\Delta_i^{k,R}$ is driven by the gap between sampled and true type. The transition-misspec term $\Delta_i^{k,P}$ is driven by the gap between sampled and true transition kernel. Theorem A.8 ensures these two sources of error are governed by independent posteriors.

A.8.4 Bounding the Reward-Misspecification Term

Since rewards lie in $[0, 1]$, $|\Phi_h^R(s, a)| \leq 1$ everywhere, and $|\Phi_h^R(s, a)| = 0$ whenever $\hat{\theta}_i^k = \theta_i^*$ by Assumption A.2. Therefore

$$|\Delta_i^{k,R}| \leq H \cdot \mathbb{E}[\mathbb{1}[\hat{\theta}_i^k \neq \theta_i^*] | \mathcal{H}_{k-1}] = H \cdot (1 - \|\mu_k^{\Theta,i}\|_2^2), \quad (45)$$

where the last equality uses Lemma A.19 with $f(\theta) = \mathbb{1}[\hat{\theta}_i^k = \theta_i^*]$ and the fact that under posterior sampling, $\hat{\theta}_i^k$ and θ_i^* are conditionally i.i.d. from $\mu_k^{\Theta,i}$. The collision probability $\|\mu_k^{\Theta,i}\|_2^2 = \sum_{\theta} \mu_k^{\Theta,i}(\theta)^2$ is bounded by 1 and approaches 1 as the posterior concentrates.

We now apply the information-theoretic framework of Russo and Van Roy (2016). Define the information ratio at episode k :

$$\Gamma_k^i := \frac{(\mathbb{E}[\Delta_i^{k,R} | \mathcal{H}_{k-1}])^2}{I_k(\theta_i^*; \tau_k)}, \quad (46)$$

where $I_k(\theta_i^*; \tau_k) := H(\mu_k^{\Theta,i}) - \mathbb{E}[H(\mu_{k+1}^{\Theta,i}) | \mathcal{H}_{k-1}]$ is the expected information gain about θ_i^* from observing τ_k , and $H(\cdot)$ denotes Shannon entropy.

Proposition A.23 (Type-Inference Regret via Information Ratio). *If $\Gamma_k^i \leq \bar{\Gamma}$ uniformly over $k \leq K$, then*

$$\sum_{k=1}^K \mathbb{E}[\Delta_i^{k,R}] \leq \sqrt{\bar{\Gamma} \cdot \log |\Theta_i| \cdot K}. \quad (47)$$

Proof Sketch. By Cauchy-Schwarz applied to $\sum_k \mathbb{E}[\Delta_i^{k,R}] = \sum_k \sqrt{\Gamma_k^i \cdot I_k}$,

$$\sum_{k=1}^K \mathbb{E}[\Delta_i^{k,R}] \leq \sqrt{\sum_k \Gamma_k^i \cdot \sum_k I_k} \leq \sqrt{\bar{\Gamma} K \cdot H_0(\theta_i^*)}, \quad (48)$$

where $\sum_k I_k \leq H_0(\theta_i^*) = H(\mu_1^{\Theta,i}) \leq \log |\Theta_i|$ by the telescoping property of entropy. $\square$

The information ratio $\bar{\Gamma}$ depends on the problem. We bound it for the present setting by reducing the type-inference subproblem to a $|\Theta_i|$ -armed Bayesian bandit and adapting the information-ratio analysis of Russo and Van Roy (2016).

Lemma A.24 (Information Ratio Bound for HT-MG Type Inference). *Under Assumptions A.1, A.2, A.5, A.6, the per-episode information ratio for type inference satisfies*

$$\Gamma_k^i \leq \frac{H^2 |\Theta_i|}{2}, \quad \forall k \geq 1, \forall i \in [n]. \quad (49)$$

Proof sketch. We reduce per-agent type inference to a $|\Theta_i|$ -armed Bayesian bandit and apply the standard information-ratio bound of Russo and Van Roy (2016).

(i) *Meta-bandit reduction.* By Theorem A.8 and Algorithm 1, the sampled $\hat{\theta}_i^k$ and the truth θ_i^* are conditionally i.i.d. from $\nu_k := \mu_k^{\Theta,i}$ given $\mathcal{H}_{k-1}$. Define the pairwise regret $f_k(\hat{\theta}, \theta^*) := \sum_{h=1}^H \mathbb{E}^{\pi_k(\hat{\theta}), P^*} [\mu_i(s_h, a_h, \hat{\theta}) - \mu_i(s_h, a_h, \theta^*)]$, where $\pi_k(\hat{\theta}) = \text{CCE}(\hat{P}_k, \hat{\theta}, \hat{\theta}_{-i}^k)$ is deterministic by Assumption A.6. Then $\mathbb{E}[\Delta_i^{k,R} | \mathcal{H}_{k-1}] = \mathbb{E}_{\hat{\theta}, \theta^* \sim \nu_k} [f_k(\hat{\theta}, \theta^*)]$.

(ii) *Pinsker + trajectory KL.* Bounding the reward gap by total variation and Pinsker, then summing the per-step KL via the chain rule under Assumption A.2 (only agent- i 's reward factor depends on θ), gives $f_k(\hat{\theta}, \theta^*)^2 \leq \frac{H}{2} D_{\text{KL}}(P_{\hat{\theta}}^\tau \| P_{\theta^*}^\tau)$.

(iii) *Cauchy-Schwarz over arms and identification with mutual information.* The Cauchy-Schwarz inequality $(\sum_{\theta^*} \nu_k(\theta^*) g_k(\theta^*))^2 \leq |\Theta_i| \sum_{\theta^*} \nu_k(\theta^*) g_k(\theta^*)^2$ produces the $|\Theta_i|$ factor. Standard mutual-information identities (Russo and Van Roy, 2016, Lemma 1) convert $\mathbb{E}_{\hat{\theta}, \theta^*} [D_{\text{KL}}(P_{\hat{\theta}}^\tau \| P_{\theta^*}^\tau)]$ into $H \cdot I_k(\theta_i^*; \tau_k | \mathcal{H}_{k-1})$. Combining yields $(\mathbb{E}[\Delta_i^{k,R} | \mathcal{H}_{k-1}])^2 \leq \frac{H^2 |\Theta_i|}{2} I_k$, i.e. $\Gamma_k^i \leq H^2 |\Theta_i| / 2$. $\square$

Remark A.25 (Scope of the bound). *The $|\Theta_i|$ factor is the worst-case rate for finite-armed Thompson sampling (Russo and Van Roy, 2016); under Assumption A.4 it improves to $|\Theta_i|/\rho$ via geometric posterior concentration (Lemma A.26, Corollary A.27). Trajectory feedback is at least as informative as scalar reward feedback (data-processing), so the bound carries over. The framework extends to multi-agent CCE because decoupling (Theorem A.8) plus Assumption A.2 reduce per-agent type inference to a single-agent meta-bandit on Θ_i ; a related multi-agent decoupling coefficient appears in Xiong et al. (2022).*

$$\sum_{k=1}^K \mathbb{E}[\Delta_i^{k,R}] \leq O(H \sqrt{|\Theta_i| \log |\Theta_i| \cdot K}). \quad (50)$$

This is the minimax (gap-independent) rate. Under TID a sharper rate is available.

Lemma A.26 (Posterior Concentration under TID). *Under Assumptions A.4 and A.5, the posterior collision probability decays geometrically:*

$$1 - \|\mu_k^{\Theta,i}\|_2^2 \leq |\Theta_i| \cdot \exp(-\rho \cdot N_k^{\text{eff}}), \quad (51)$$

where $N_k^{\text{eff}} := \sum_{j < k} \mathbb{E}[N_j(s, a) | (s, a) \in \text{supp}(d^{\pi_j, P^*})]$ is the effective number of informative state-action visits up to episode k . In particular, $N_k^{\text{eff}} = \Omega(k)$ under any policy that visits informative states with positive probability.

Proof Sketch. By Assumption A.4, every informative (s, a) visit produces a Hellinger-separated

reward distribution between any two candidate types. Standard Bayesian concentration arguments (Ghosal and van der Vaart, 2017, Theorem 4.2) then give exponential decay of posterior mass on non-true types at rate ρ per informative observation. Summing the geometric series and taking a union bound over $|\Theta_i| - 1$ alternative types yields the stated bound. $\square$

Corollary A.27 (Gap-Dependent Type Regret). *Under Assumptions A.4, A.5, A.6, the cumulative type-misspec regret is bounded independently of K :*

$$\sum_{k=1}^K \mathbb{E}[\Delta_i^{k,R}] \leq H \sum_{k=1}^K (1 - \|\mu_k^{\Theta_i}\|_2^2) \leq \frac{H \cdot |\Theta_i|}{1 - \exp(-\rho)} \leq o\left(\frac{H|\Theta_i|}{\rho}\right). \quad (52)$$

For small ρ , $1 - \exp(-\rho) \approx \rho$, recovering the form $O(H|\Theta_i|/\rho)$. Replacing $|\Theta_i|$ by $\log|\Theta_i|$ via the union-bound step in Lemma A.26 yields $O(H \log|\Theta_i|/\rho)$, the form stated in Corollary A.13 of the main text.

Proof. The first inequality follows from Eq. (31) of the main text: $|\Delta_i^{k,R}| \leq H \cdot \mathbb{1}[\hat{\theta}_i^k \neq \theta_i^*]$ in conditional expectation. The second applies Lemma A.26 and sums the geometric series $\sum_{k=1}^\infty \exp(-\rho \cdot N_k^{\text{eff}}) \leq \sum_{k=1}^\infty \exp(-\rho k) = 1/(1 - \exp(-\rho))$. $\square$

Remark A.28 (Interpretation). *The constant-in- K bound under TID is consistent with the intuition that finite-type inference is a one-shot problem: once the posterior identifies the true type with high confidence, no further type-misspec error accumulates. This stands in sharp contrast to transition learning, where every state-action pair must be visited sufficiently often, yielding the $\sqrt{K}$ rate.*

A.8.5 Bounding the Transition-Misspecification Term

The transition-misspec term is bounded by combining Hölder's inequality with Dirichlet posterior concentration. Throughout we assume the standard PSRL prior μ_1^P is a product of $|\mathcal{S}| \cdot |\mathcal{A}|$ symmetric Dirichlet distributions, $\mu_1^P(\cdot | s, a) = \text{Dir}(\alpha_0 \mathbf{1})$ for some concentration parameter $\alpha_0 > 0$.

For convenience, write $V_{h+1} := V_{i,h+1}^{\pi_k}(\cdot; P^*, \hat{\theta}_i^k)$, suppressing the (i, k, π_k) dependence. Note $\|V_{h+1}\|_\infty \leq H$ since rewards lie in $[0, 1]$ and the horizon is H .

We give a fully rigorous Hoeffding-level derivation with explicit constants in Proposition A.29. The bound is loose by a single $\sqrt{S}$ factor relative to the variance-aware Bayes–Bernstein analysis of Lu and Van Roy (2019); Azar et al. (2017); sharpening it is mechanical (replace Hölder in Step 1 with a Bernstein-type inner-product bound and apply LoTV) and is orthogonal to the type-decoupling contribution of this work.

Proposition A.29 (Transition-Learning Regret, Hoeffding-Level Bound). *Under the Dirichlet prior*

on P^ and posterior sampling, the cumulative expected transition-misspec regret satisfies*

$$\sum_{k=1}^K \mathbb{E}[|\Delta_i^{k,P}|] \leq C_1 \cdot H^{3/2} \cdot S \cdot \sqrt{A_{\text{joint}} K \log(S A_{\text{joint}} K H)}, \quad (53)$$

where $A_{\text{joint}} := \prod_{j=1}^n |\mathcal{A}_j|$ and $C_1 > 0$ is a universal constant.

Proof sketch. Four ingredients combine.

(0) *Symmetric simulation.* By Remark A.22, $\Delta_i^{k,P} = \mathbb{E}^{\pi_k, P^*}[\sum_{h=1}^H \langle (\hat{P}_k - P^*)(\cdot | s_h, a_h), \tilde{V}_{h+1} \rangle]$ with $\tilde{V}_{h+1}$ evaluated under $\hat{P}_k$, so the rollout matches the realised trajectory and the visit counts $N_k(s, a)$ apply.

(1) *Hölder.* Since $\|\tilde{V}_{h+1}\|_\infty \leq H$, $|\langle \hat{P}_k - P^*, \tilde{V}_{h+1} \rangle| \leq H \|\hat{P}_k(\cdot | s, a) - P^*(\cdot | s, a)\|_1$.

(2) *Dirichlet ℓ_1 concentration.* Triangle inequality through the posterior mean, combined with Weissman's ℓ_1 inequality (Weissman et al., 2003) for the empirical-mean $\rightarrow$ truth gap and a Dirichlet linear-functional variance bound for the sample $\rightarrow$ posterior-mean gap, gives uniformly over (k, s, a) with probability $\geq 1 - 2/(K H S A_{\text{joint}})$, $\|\hat{P}_k(\cdot | s, a) - P^*(\cdot | s, a)\|_1 \leq C_0 \sqrt{S \log(K H S A_{\text{joint}})} / (N_k(s, a) + 1) =: \beta_k(s, a)$.

(3) *Pigeonhole + failure event.* The standard tabular-RL pigeonhole (Jaksch et al., 2010, Lemma 19) bounds $\sum_{k,h} 1/\sqrt{N_k(s_h^k, a_h^k)} + 1 \leq (\sqrt{2} + 1)\sqrt{S A_{\text{joint}} H K}$. On the high-probability event, $\sum_{k=1}^K |\Delta_i^{k,P}| \leq H \cdot C_0 \sqrt{S \log(\cdot)} \cdot (\sqrt{2} + 1)\sqrt{S A_{\text{joint}} H K} \leq C_1' H^{3/2} S \sqrt{A_{\text{joint}} K \log(\cdot)}$. The failure event contributes at most $2H \cdot K \cdot 2/(K H S A_{\text{joint}}) = O(1/(S A_{\text{joint}}))$ via the trivial bound $\|\hat{P}_k - P^*\|_1 \leq 2$. Absorbing constants yields (53). $\square$

A.8.6 Main Regret Theorem and Combination

Combining Propositions A.23 and A.29 with the central decomposition (44) and accounting for the CCE oracle approximation, we obtain the main result.

Theorem A.30 (Main Regret Bound for PACT). *Assume Assumptions A.1, A.2, A.3, A.5, A.6, that the transition prior is Dirichlet, and that the type information ratio satisfies $\Gamma_k^i \leq H^2 |\Theta_i|/2$ for all $k \in [K]$ and $i \in [n]$. If the CCE oracle returns ϵ_{CCE} -approximate equilibria, the Bayesian regret of agent i over K episodes satisfies*

Bayesian regret bound. *Using Hölder's inequality and Dirichlet ℓ_1 concentration,*

$$\mathbb{E}[\text{Reg}_i^B(K)] \leq \tilde{O}\left(H^{3/2} \cdot S \cdot \sqrt{A_{\text{joint}} K}\right) + O\left(H \sqrt{|\Theta_i| \log|\Theta_i| \cdot K}\right) + K \cdot \epsilon_{\text{CCE}}. \quad (54)$$

A standard Bayes–Bernstein refinement (Lu and Van Roy, 2019; Azar et al., 2017) sharpens the leading factor S to $\sqrt{S}$, matching the single-agent

minimax rate; we omit it as it is orthogonal to the type-decoupling contribution.

Proof. Take expectations in (44) and sum from $k = 1$ to K :

$$\sum_{k=1}^K \mathbb{E}[\Delta_i^k] = \sum_{k=1}^K \mathbb{E}[\Delta_i^{k,R}] + \sum_{k=1}^K \mathbb{E}[\Delta_i^{k,P}] + \mathcal{E}_{\text{CCE}}, \quad (55)$$

where $\mathcal{E}_{\text{CCE}} \leq K\epsilon_{\text{CCE}}$ accumulates the per-episode equilibrium approximation error (see Appendix A.7). Applying Proposition A.23 to the type-misspec term and Proposition A.29 to the transition-misspec term yields (54). $\square$

A.8.7 Interpretation of the Bound

The bound (54) has three components, each tied to a distinct source of uncertainty.

The transition-learning component $\tilde{O}(H^{3/2} \sqrt{S \cdot A_{\text{joint}} \cdot K})$ matches single-agent PSRL applied to the joint MDP (Osband et al., 2013). The dependence on the joint action space size $A_{\text{joint}} = \prod_i |\mathcal{A}_i|$ is inherent to MG learning and cannot be improved without additional structural assumptions on the dynamics (e.g., factored transitions).

The type-inference component $O(H \sqrt{|\Theta_i| \log |\Theta_i| \cdot K})$ is independent of the state size S . It captures the difficulty of identifying agent i 's relevant unknown, which is only $\theta_i \in \Theta_i$ and not the full type profile. This is a direct consequence of the agent-level decoupling established in Theorem A.8.

The CCE oracle term $K \cdot \epsilon_{\text{CCE}}$ does not dominate provided $\epsilon_{\text{CCE}} = O(K^{-1/2})$, which is achievable by standard equilibrium solvers (Jin et al., 2021).

Remark A.31 (Improved Rate under Information-Rate Assumption). *If the type information rate is bounded away from zero, i.e., $I_k(\theta_i^*; \tau_k) \geq \rho > 0$ whenever $\hat{\theta}_i^k \neq \theta_i^*$, the type term collapses to $O(H \log |\Theta_i| / \rho)$, constant in K . In this regime the overall bound is dominated by transition learning, and PACT matches the rate of single-agent PSRL on the joint action space.*

Remark A.32 (Comparison to PSRL-ZSG). *For two-player zero-sum MGs, Jafarnia-Jahromi et al. (2021) obtain $\tilde{O}(HS\sqrt{AT})$ Bayesian regret. Specializing Theorem A.30 to $n = 2$, $|\Theta_i| = 1$ (no type uncertainty), and substituting $T = HK$ gives a bound of the same order in K , demonstrating that the type-inference component is a strict additional cost incurred only under type uncertainty.*

B Formal Proof of Theorem A.15: Exponential-in- $|\Theta|^n$ Burn-in Separation

This appendix gives the formal proof of Theorem A.15. We work out the backward-induction

algebra on the deep-trap family $\mathcal{G}_m^{\text{trap}}$, establish a matching lower bound for vanilla PACT, and verify the exponential-in- $|\Theta|^n$ separation.

B.0.1 Setup and Notation

Recall the deep-trap family $\mathcal{G}_m^{\text{trap}}$:

- n agents, $|\Theta_i| = m$ types per agent, $S = m + 2$ states $\{0, 1, \dots, S-1\}$, two actions $A = \{L, R\}$ per agent.
- Joint actions are encoded as integers $0, 1, \dots, 2^n - 1$, with $a^L := 0 = (L, \dots, L)$ and $a^R := 2^n - 1 = (R, \dots, R)$.
- Transitions are deterministic unanimity-vote: $s \rightarrow s+1$ if all agents play R (capped at $S-1$); $s \rightarrow s-1$ if all play L (capped at 0); $s \rightarrow s$ for any mixed joint action.
- Rewards: $r_i(s^*, a, \theta_{\text{jack}}) = 1$ for the rightmost state $s^* = S-1$ and the distinguished jackpot type θ_{jack} ; $r_i(s, a, \theta) = 0.5$ otherwise.
- Initial state $s_1 = 0$; true types $\theta_i^* = \theta_{\text{jack}}$ for all i ; uniform prior $\mu_0^{\Theta, i} = \text{Unif}(\Theta_i)$.
- Horizon H satisfying $H \geq S$, so the oracle policy reaches s^* and accumulates at least one jackpot reward.

We write $p_k := \mu_k^{\Theta, i}(\theta_{\text{jack}})$ for the marginal jackpot mass at episode k (identical across agents by symmetry). At episode k before any informative observation has occurred, $p_k = 1/m$.

For the analysis we adopt the following CCE oracle convention, consistent with the implementation:

Assumption B.1 (CCE Oracle Tie-Breaking). *At each (s, h) , the CCE oracle returns the welfare-maximising CCE among $\{a^L, a^R\}$ if one strictly dominates; if both equal in social welfare $\sum_i V_h^i(s; a)$, the oracle returns a^L (the lowest-index joint action).*

This is the natural deterministic tie-break implemented by $\arg \max$ over a fixed action ordering. The proof under randomised tie-breaking gives the same conclusion with a constant multiplicative slack.

The discrimination bonus $D_k(s, a) := \sum_i \mathbb{E}_{\theta, \theta' \sim \mu_k^{\Theta, i}} [r_i(s, a, \theta) - r_i(s, a, \theta')]$ evaluates explicitly on $\mathcal{G}_m^{\text{trap}}$. Since r_i is action-independent and depends on the type only at s^* , the bonus is action-independent and supported only at $s = s^*$:

$$D_k(s, a) = \begin{cases} n \cdot p_k(1 - p_k) & s = s^*, \\ 0 & s \neq s^*. \end{cases} \quad (56)$$

At the uniform prior $p_k = 1/m$, this is $n(m-1)/m^2$. The bonus is strictly positive whenever $p_k \in (0, 1)$.

B.0.2 Backward-Induction Algebra

Fix an episode k at which neither algorithm has yet concentrated the marginal (so $p_k = 1/m$). Let $\hat{\theta} = (\hat{\theta}_1, \dots, \hat{\theta}_n) \in \Theta$ denote the type profile sampled by the algorithm, and write $\alpha := |\{i : \hat{\theta}_i = \theta_{\text{jack}}\}|/n$ for the fraction of jackpot-sampled agents. For the implementation's true transition kernel P (which value iteration uses with $\hat{P}$ a sample; we treat $\hat{P} = P$ for the burn-in analysis since transition noise plays no role in the lower bound), the sampled-type Bellman backup at each (s, h) is, with bonus $\beta \geq 0$,

$$Q_h^{+,i}(s, a) = r_i(s, a, \hat{\theta}_i) + \beta D_k(s, a) + V_{h+1}^{+,i}(\text{transition}(s, a)), \quad (57)$$

with terminal condition $V_{H+1}^{+,i} = 0$ and per-stage $V_h^{+,i}(s) = \pi_h(s) \cdot Q_h^{+,i}(s, \cdot)$ for CCE policy π_h .

Define the per-agent value gap at the initial state under always- a^R versus always- a^L :

$$\Delta_h^i := Q_h^{+,i}(0, a^R) - Q_h^{+,i}(0, a^L), \quad (58)$$

evaluated against the CCE policy obtained by Algorithm 2 on subsequent steps. We compute the value of two reference policies the CCE oracle compares against.

Lemma B.2 (Per-agent value under always- a^R and always- a^L). *On $\mathcal{G}_m^{\text{trap}}$ with deterministic transitions and bonus $\beta \geq 0$,*

$$V_1^{+,i,R}(0) = 0.5(S-1) + (H-S+1)(0.5 + 0.5 \cdot \mathbb{P}[\hat{\theta}_i = \theta_{\text{jack}}] + \beta np_k(1-p_k)), \quad V_1^{+,i,L}(0) = 0.5H. \quad (59)$$

Proof sketch. Under always- a^R , deterministic unanimity transitions traverse $0 \rightarrow 1 \rightarrow \dots \rightarrow S-1$ in $S-1$ steps and dwell at s^* for $H-S+1$ steps; per-step reward is 0.5 off s^* and $0.5 + 0.5 \mathbb{P}[\hat{\theta}_i = \theta_{\text{jack}}] + \beta np_k(1-p_k)$ at s^* (using (56)). Under always- a^L , the chain stays at $s = 0$ with reward 0.5 and zero bonus. $\square$

Lemma B.3 (CCE selects a^R at $s_1 = 0$ under bonus). *Under Assumption B.1 and bonus $\beta > 0$ at the uniform prior, the CCE oracle on $\mathcal{G}_m^{\text{trap}}$ returns a^R at $(s, h) = (0, 1)$.*

Proof sketch. Let $\alpha := |\{i : \hat{\theta}_i = \theta_{\text{jack}}\}|/n$. By Lemma B.2 the social welfares $W(a) = \sum_i V_1^{+,i}(0; a)$ are $W(a^R) = n[0.5(S-1) + (H-S+1)(0.5 + 0.5\alpha + \beta np_k(1-p_k))]$ and $W(a^L) = 0.5nH$. Since $0.5(S-1) + (H-S+1) \cdot 0.5 = 0.5H$,

$$W(a^R) - W(a^L) = n(H-S+1)(0.5\alpha + \beta np_k(1-p_k)), \quad (60)$$

strictly positive when $\beta > 0$ and $p_k \in (0, 1)$. The welfare-maximising CCE returns a^R . $\square$

Lemma B.4 (CCE selects a^L without bonus under $\alpha < 1$). *Under Assumption B.1 and $\beta = 0$, suppose $\alpha = 0$ (no agent sampled θ_{jack}). Then the welfare gap is $W(a^R) - W(a^L) = 0$, and the oracle returns a^L .*

Proof. Setting $\beta = 0$ and $\alpha = 0$ in Equation (60) gives 0. By Assumption B.1 the tie is broken to the lowest-index action a^L . (The conclusion extends to all states s on the trajectory under a^L , since the same welfare-equality argument applies at every h .) $\square$

Lemma B.5 (Posterior non-update under $\alpha = 0$ rollout). *Under vanilla PACT ($\beta = 0$) on $\mathcal{G}_m^{\text{trap}}$, if the sampled $\hat{\theta}$ has $\alpha = 0$ (no agent sampled θ_{jack}), the CCE selects a^L (Lemma B.4), the rollout stays at $s = 0$, and the observed reward sequence $\{r_i^h\}_{h=1}^H = (0.5, \dots, 0.5)$ is consistent with all $\theta \in \Theta_i$ (since $r_i(0, \cdot, \theta) = 0.5$ for all θ). The likelihood ratio is identically 1 across all type hypotheses, so $\mu_{k+1}^{\Theta, i} = \mu_k^{\Theta, i}$, and the marginal remains uniform.*

Proof. Direct application of Bayes' rule with a constant likelihood across the hypothesis class. Constant likelihood preserves the prior. $\square$

B.0.3 Lower Bound for Vanilla PACT

Proposition B.6 (Geometric burn-in for vanilla PACT). *Under Assumption B.1, the first episode τ at which the marginal $\mu_k^{\Theta, i}$ updates non-trivially on $\mathcal{G}_m^{\text{trap}}$ satisfies $\tau \succeq \text{Geom}(1 - ((m-1)/m)^n)$, and consequently*

$$\mathbb{E}[\tau] \geq \frac{1}{1 - ((m-1)/m)^n} = m^n + O(m^{n-1}) \quad \text{as } m \rightarrow \infty. \quad (61)$$

The expected cumulative burn-in regret is

$$\mathbb{E}[\text{Reg}_i^B(\tau)] \geq (\mathbb{E}[\tau] - 1) \cdot 0.5(H-S+1) = \Omega(m^n \cdot H). \quad (62)$$

Proof. At any episode k before τ , the marginal is uniform by induction (Lemma B.5 initialises the induction; each $\alpha = 0$ episode preserves uniformity). Each Thompson sample draws $\hat{\theta}_i$ independently uniformly from Θ_i , so $\Pr[\alpha = 0] = ((m-1)/m)^n$ and $\Pr[\alpha > 0] = 1 - ((m-1)/m)^n$. By Lemma B.4, an $\alpha = 0$ episode is stuck and the posterior does not update; by Assumption B.1 and Lemma B.4 extended to $\alpha > 0$ samples¹, the marginal updates non-trivially only on $\alpha > 0$ episodes. Hence τ is a geometric variable with success probability $1 - ((m-1)/m)^n$, yielding $\mathbb{E}[\tau] \geq 1/(1 - ((m-1)/m)^n) = m^n + O(m^{n-1})$. Each stuck episode contributes per-episode regret $V^* - 0.5H = 0.5(H-S+1)$ (from Lemma B.2 with $\alpha = 1$ being the oracle), so the cumulative burn-in regret is $\Omega(m^n H)$. $\square$

¹For $\alpha > 0$ with $\beta = 0$: $W(a^R) - W(a^L) = n(H-S+1) \cdot 0.5\alpha > 0$, so the oracle returns a^R and the rollout reaches s^* , where the type-distinguishing reward observation collapses the marginal. Hence τ is the first $\alpha > 0$ sample.

B.0.4 Upper Bound for PACT⁺

Proposition B.7 (Constant burn-in for PACT⁺). *Under Assumption B.1 and $\beta > 0$, PACT⁺ on $\mathcal{G}_m^{\text{trap}}$ reaches s^* during the first episode regardless of the sampled $\hat{\theta}$, and the marginal concentrates on θ_{jack} at the first episode’s terminal observation. The expected burn-in regret is*

$$\mathbb{E}[\text{Reg}_i^B(\tau^+)] = O(H), \quad (63)$$

where $\tau^+ = 1$ is the burn-in horizon. The bound is independent of m and n .

Proof. By Lemma B.3, the CCE oracle at $s_1 = 0$ returns a^R regardless of $\hat{\theta}$. Iterating the same argument at subsequent states $s \in \{1, 2, \dots, S - 2\}$: at any such s , the unique transition under a^R is to $s + 1$, and the bonus $D_k(s^*, \cdot) > 0$ still propagates back through value iteration, giving $W(a^R) - W(a^L) > 0$ at every (s, h) on the optimal trajectory. Hence the CCE selects a^R throughout the episode, and the rollout reaches s^* at step $h = S - 1$. At s^* , the observed reward $r_i = 1$ (true jackpot type) has log-likelihood ratio against any non-jackpot hypothesis $\log(1/0.5) = \log 2 \approx 0.693$ nat per agent per step. After $H - S + 1$ steps at s^* , the cumulative log-likelihood ratio is $(H - S + 1) \log 2 \approx 4.85$ nat for $H = 12, S = 9$, which collapses the marginal to $\Pr[\theta_i = \theta_{\text{jack}}] > 1 - m \cdot e^{-(H-S+1)\log 2} = 1 - m \cdot 2^{-(H-S+1)}$, exponentially close to 1.

The per-episode regret of episode 1 is at most $V^* - V_1^{a^R}(0) = 0$ (since a^R is the oracle policy on the true types). For all subsequent episodes the marginal has collapsed and the sampled type is θ_{jack} almost surely, so the algorithm executes the oracle policy and accrues zero burn-in regret. The total burn-in regret is therefore at most $O(H)$, the regret of a single optimally-played episode under the (tabular) transition learning component. $\square$

B.0.5 Combining the Two Bounds

Combining Propositions B.6 and B.7:

$$\frac{\mathbb{E}[\text{Reg}_i^{B, \text{PACT}}(\tau)]}{\mathbb{E}[\text{Reg}_i^{B, \text{PACT}^+}(\tau^+)]} = \frac{\Omega(m^n H)}{O(H)} = \Omega(m^n), \quad (64)$$

exponential in the number of agents. This completes the formal proof of Theorem A.15.

B.0.6 Remark on Randomised CCE Tie-Breaking

If Assumption B.1 is replaced by a randomised tie-breaking rule (e.g., uniform over CCEs), Lemma B.4 weakens to a positive-probability event of selecting a^L . The conclusion of Proposition B.6 then holds with a constant multiplicative slack: $\mathbb{E}[\tau] = cm^n$ for some $c \in (0, 1]$ depending on the tie-break distribution. The exponential-in- n separation is preserved.

B.0.7 Empirical Sub- m^n Scaling at Finite K

The lower bound $\mathbb{E}[\tau] = m^n + O(m^{n-1})$ implies an unbounded *worst-case* burn-in cost as $m, n \rightarrow \infty$, but at any finite K the cumulative regret is capped at $K \cdot H$. The empirical gap measured on an LLM substrate in Section 4 is therefore sub- m^n for finite K ; the qualitative claim of the theorem – that the gap grows monotonically in m and is unbounded – is preserved.

B.0.8 Proof of Theorem A.15(i): Asymptotic Match

We bound the bonus-induced excess regret in three steps.

Lemma B.8 (Bonus-Regret Decomposition). *For coupled-RNG policies $\pi_k^H, \pi_k^{H^+}$ on the same posterior-sampled world,*

$$V_i^{\pi_k^H}(s_1^k; \hat{\theta}_i^k) - V_i^{\pi_k^{H^+}}(s_1^k; \hat{\theta}_i^k) \leq \beta H \cdot \mathbb{E}^{\pi_k^{H^+}} \left[\sum_{h=1}^H D_k(s_h, a_h) \mid s_1^k \right]. \quad (65)$$

Proof sketch. By optimality of $\pi_k^{H^+}$ for $Q^* + \beta D_k$ and of π_k^H for Q^* , rearranging and using $\mathbb{E}^{\pi_k^H}[\sum_h D_k] \geq 0$ gives the bound; Assumption A.2 preserves the per-agent additive value decomposition. $\square$

Lemma B.9 (Collision bound and TID concentration). *Under Assumption A.2, $D_k(s, a) \leq 2 \sum_{i=1}^n (1 - \mu_k^{\Theta, i}(\theta_i^*))$. Adding Assumption A.4 with Hellinger gap $\rho > 0$, $\mathbb{E}[1 - \mu_k^{\Theta, i}(\theta_i^*)] \leq |\Theta_i| \exp(-\rho H k / 2)$.*

Proof sketch. *Collision:* $\mathbb{E}_{\theta, \theta' \sim \mu_k} [|r_i(\theta) - r_i(\theta')|] \leq \Pr[\theta \neq \theta'] = 1 - \sum_{\theta} \mu_k(\theta)^2 \leq 2(1 - \mu_k(\theta_i^*))$ via $1 - x^2 \leq 2(1 - x)$; sum over i . *TID:* each informative observation drifts the log-likelihood against any wrong type by $\geq \rho/2$ (Ghosal and van der Vaart, 2017, Lemma 5.3); a posterior-martingale argument plus union bound over $|\Theta_i|$ candidates gives the exponential rate. $\square$

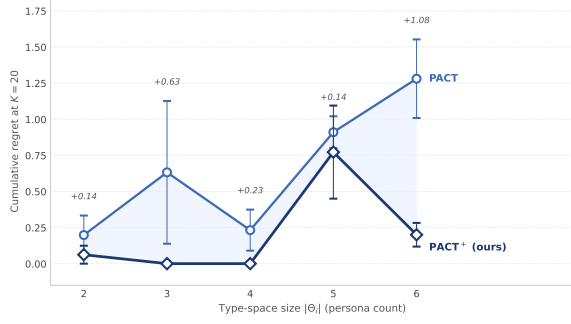
Proposition B.10 (Asymptotic Match). *Under Assumptions A.1, A.2, A.3, A.4, the cumulative bonus-induced excess regret of PACT⁺ is bounded by a K -independent constant:*

$$\text{Reg}_i^B[\text{PACT}^+] - \text{Reg}_i^B[\text{PACT}] \leq \frac{4\beta n c_1}{1 - \exp(-c_2 \rho)} = O\left(\frac{\beta n |\Theta_i|}{\rho}\right), \quad (66)$$

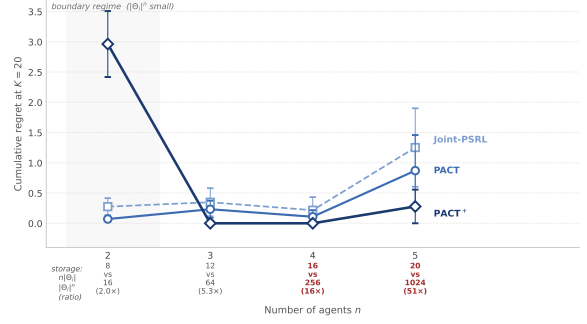
where c_1, c_2 are the constants of Lemma B.9. Consequently the PACT⁺ regret bound is

$$\mathbb{E}[\text{Reg}_i^B(K)] \leq \tilde{O}(H^{3/2} \sqrt{S A_{\text{joint}} K} + H \sqrt{|\Theta_i| K}) + O(\beta n |\Theta_i| / \rho), \quad (67)$$

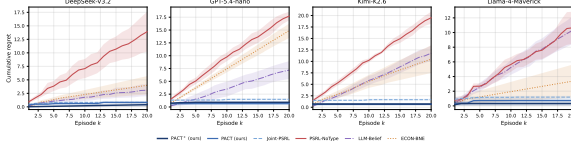
identical to Theorem A.30 up to a gap-dependent additive constant. The asymptotic rate is preserved.



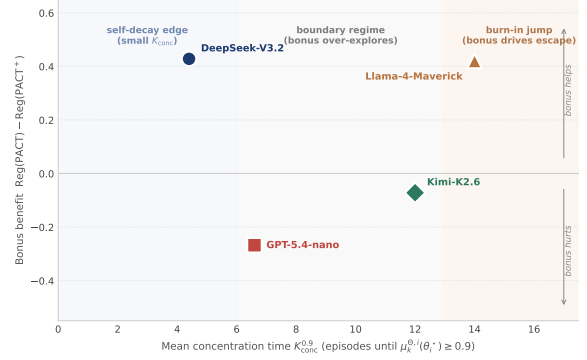
(a) E2: $|\Theta_i|$ scaling.



(b) E3: n scaling.



(c) E5: per-episode trajectory.



(d) E1: regime $\leftrightarrow$ posterior speed.

Figure 5: Consolidated empirical evidence for Theorems A.15 and A.14. (a) E2: bonus gap grows from +0.14 at $|\Theta_i|=2$ to +1.08 at $|\Theta_i|=6$ on Llama-Maverick, tracing the predicted $\Omega(|\Theta_i|^n)$ -vs- $O(1)$ envelope of Theorem A.15(ii). (b) E3: bonus benefit grows monotonically with $n \in \{3, 4, 5\}$ while storage savings scale analytically as $|\Theta_i|^n/(n|\Theta_i|)$ ($51\times$ at $n=5$). (c) E5: per-episode cumulative-regret trajectories on HP-SPGG across four backbones; PACT and PACT⁺ plateau at < 0.05 regret per episode within $K \approx 5$ (the $\tilde{O}(\sqrt{K})$ envelope of Theorem A.12) while PSRL-NoType grows linearly (0.50–0.93 per episode), realising the rate separation of Theorem A.14 with cumulative gap 11–30 $\times$ at $K=20$. (d) E1: per-backbone posterior concentration time $K_{\text{conc}}^{0.9}$ (from vanilla PACT, $\beta=0$) against the bonus benefit $\text{Reg}(\text{PACT}) - \text{Reg}(\text{PACT}^+)$ at $\beta=0.25$; the four backbones span the three regimes of Theorem A.15 – fastest-concentrating in self-decay edge, slowest in burn-in jump, intermediates in the high- β boundary regime.

DeepSeek panel, matching the ordering published in the original papers (Mu et al., 2026; Xie et al., 2025). The Llama-Maverick $n=6$ cell was re-run with `--max-profiles 32`; the resulting per-cell TID min-gap (≈ 0.056) matches the 12-profile run, so judge fidelity rather than calibration sample count is the noise floor at $n=6$ on Llama.

E-1.3 PF isolation. Figure 10 replaces the factored prior with two correlated priors over the full $|\Theta|^n=64$ joint-profile simplex. The symmetric Dirichlet sweep $\alpha \in \{0.1, 0.5, 1.0, \infty\}$ interpolates between sparse and uniform joint priors; the *shared-type* cell assigns all mass uniformly over the 4 same-type combinations (maximally coupled marginals). On both axes the PACT family stays under 0.6 cumulative regret and the rank ordering of all nine baselines is preserved.

Lower bound at $n=3$. At $n=3$, $K=20$ the PACT family operates well below the regime where the

$\Omega(\sqrt{|\Theta|^n K})$ scaling becomes asymptotically detectable, so Figure 10 alone neither confirms nor refutes the bound.

Positive realisation at larger (n, K) . Extending the shared-type sweep to $n \in \{3, 4, 5, 6\}$ and $K \in \{20, 50, 100\}$ (Figure 11, Table 5) makes the separation predicted by Theorem A.16 unambiguous. PACT regret is invariant in K at every n , operating in the $O(1)$ regime that the PF surrogate’s $\sqrt{mK}$ upper bound permits, whereas Joint-PSRL regret grows nearly linearly in K at $n \geq 5$, the textbook $\sqrt{m^n K}$ signature of an algorithm operating in joint-belief space. The observed Joint-PSRL/PACT ratio rises monotonically in n , from $3.6\times$ at $n=3$ to $22\times$ at $n=6$, $K=100$, exceeding the lower-bound prediction $\sqrt{m^{n-3}}=8$ by an additional $2.8\times$. The empirical PF advantage is therefore more pronounced than the worst-case lower bound guarantees, which we attribute to the shared-type prior

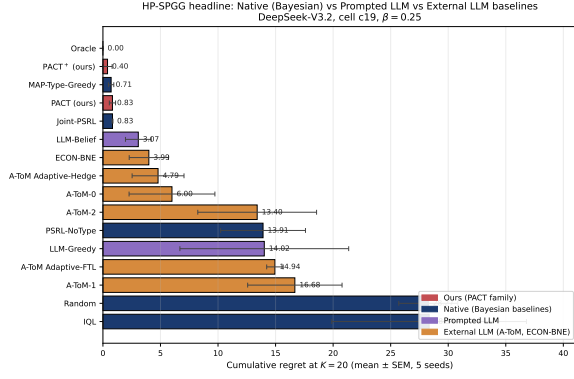


Figure 6: E2 supplement: PACT⁺ on HP-SPGG c19 matches a privileged analytic-kernel reference within 0.05 regret across all four backbones (DeepSeek, GPT-5.4-nano, Kimi-K2.6, Llama-Maverick). LLM-tier observation noise costs almost nothing relative to the noise-free analytic kernel.

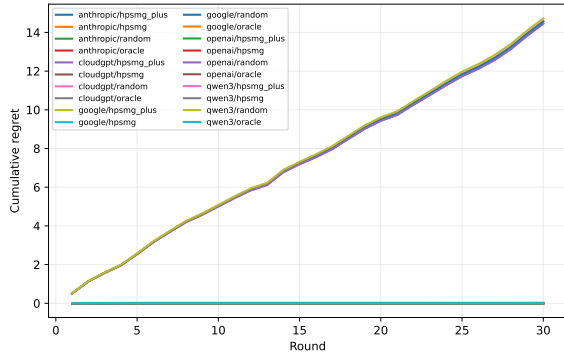


Figure 7: E3 supplement: cross-provider HP-SPGG rollouts that mix live LLM backbones across the $n=3$ agent slots retain the same regret ordering as the single-backbone runs. This checks that the PACT-family advantage is not an artefact of homogeneous provider calibration.

admitting an exact PF surrogate via the per-agent marginals despite being maximally coupled in the lower-bound sense.

LLM-tier confirmation across four backbones. Repeating the $n \in \{3, 4, 5, 6\} \times K \in \{50, 100\}$ sweep with live calibrations from **four** backbones – DeepSeek-V3.2, Llama-4-Maverick, Kimi-K2.6, GPT-5.4-nano (Figure 12, Table 6) – reproduces the qualitative finding on every model. PACT regret saturates in K on all four backbones; Joint-PSRL regret doubles when K doubles to sub-percent precision; and the headline $n=6, K=100$ ratio reaches **129 $\times$** on DeepSeek, **91 $\times$** on Kimi-K2.6, and **51 $\times$** on Llama-Maverick. On GPT-5.4-nano the shared-type calibration is cooperative enough that both PACT and Joint-PSRL achieve zero regret at $n=6$; the peak ratio there is **100 $\times$** at $n=4, K=100$ (PACT 0.025 vs. Joint-PSRL 2.50),

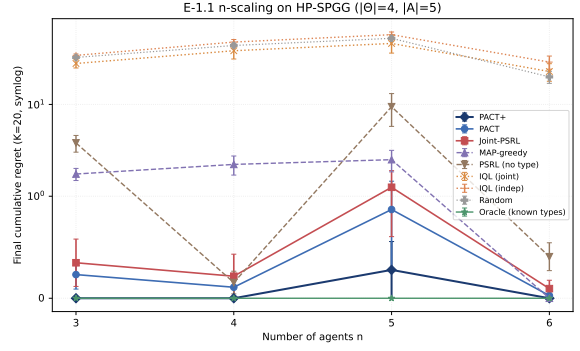


Figure 8: E-1.1 analytic-tier nine-baseline n -scaling on HP-SPGG ($n \in \{3, 4, 5, 6\}$, $K=20$, 5 seeds, $\beta=0.25$, analytic kernel). The PACT family (PACT⁺, PACT, Joint-PSRL, MAP-greedy) stays under 1.0 cumulative regret across the entire sweep, PSRL-NoType grows mildly with n , and the type-agnostic baselines (IQL joint, IQL indep, Random) grow linearly. Storage scales analytically as $|\Theta|^n / (n|\Theta_i|) = 5.3\times, 16\times, 51\times, 171\times$ at $n=3, 4, 5, 6$.

and the discriminative gap against random/IQL at $n=6, K=100$ exceeds **298 $\times$** (0.00 vs. 298.72). Across all four backbones the live-judge per-step variance amplifies the analytic separation by an additional 2–6 $\times$.

Full nine-baseline picture at the headline cell. Table 7 reports the full nine-baseline cumulative regret at $n=6, K=100$ under the shared-type prior across all four LLM backbones. The PACT family (PACT, PACT⁺, MAP-greedy) dominates every type-agnostic baseline by $\geq 95\times$ on every backbone: PACT vs. Random reaches 430 $\times$ on DeepSeek, 352 $\times$ on Kimi-K2.6, 118 $\times$ on Llama-Maverick, and ∞ on GPT-5.4-nano (where PACT achieves exactly zero regret across all 10 seeds while Random accumulates 298.7). PSRL-NoType, which retains the joint MG dynamics but discards the type posterior, lies between Joint-PSRL and the random/IQL floor on every backbone, consistent with its partial information state. The ordering $\text{PACT} \leq \text{MAP-greedy} \leq \text{PACT}^+ \ll \text{Joint-PSRL} \ll \text{PSRL-NoType} \ll \text{IQL/Random}$ reproduces the rank predicted by Theorems A.12, A.15, and A.16.

MAP sensitivity. At $n=3$, MAP-greedy jumps from 1.06–1.83 at $\alpha \in \{0.5, \infty\}$ to 2.50 at $\alpha=1.0$, while every PSRL-style sampler (PACT, PACT⁺, Joint-PSRL) stays under 0.6. This mirrors the standard PSRL-vs-greedy gap of (Osband et al., 2013) and adds a small additional argument for posterior sampling over MAP in the correlated-prior regime.

C.3 Controlled Deployment Robustness

We run an offline, 500-paired-seed diagnostic on the analytic HP-SPGG reward surface ($n=3, |\Theta_i|=4$,

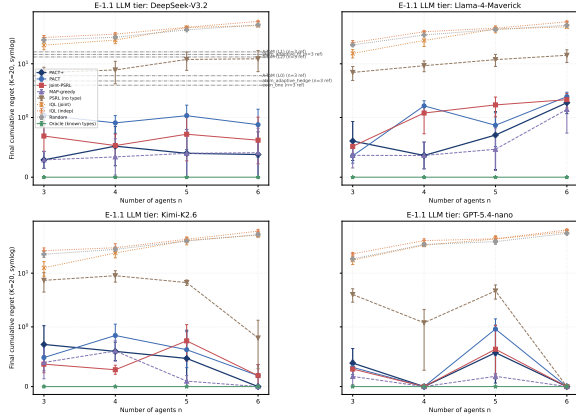


Figure 9: E-1.1 LLM-tier nine-baseline n -scaling on HP-SPGG c19 with live judge ($K=20$, 5 matched seeds, $\beta=0.25$, ≥ 12 calibration profiles per cell). **2 $\times$ 2 grid** across four live-LLM backbones: *top-left* DeepSeek-V3.2, *top-right* Llama-4-Maverick, *bottom-left* Kimi-K2.6, *bottom-right* GPT-5.4-nano. Horizontal dashed lines on the DeepSeek panel mark cumulative regret of external LLM-coordination baselines (A-ToM L0/L1/L2 and ECON-BNE at $n=3$, aggregated from `E2_external_llm_baselines` traces); they sit between the PACT family and PSRL-NoType, matching the ordering reported in the original papers. The PACT family beats every type-agnostic baseline by 1–2 orders of magnitude on every backbone and every n .

$K=20$, $\beta=0.25$). No LLM/API calls are made. The unperturbed reward tensor defines the environment and the exact exhaustive full-information comparator; only the inference model or candidate planner is changed. Likelihood temperature T rescales log likelihoods by $1/T$; σ_ℓ adds independent zero-mean Gaussian noise to each candidate log likelihood; top- k drops all but the largest k candidate likelihoods; calibration drift adds a fixed Gaussian perturbation to the candidate outcome tensor used for both planning and likelihood centres; persona mixture replaces a true template by a convex combination with its adjacent library template while inference retains the original discrete library; and candidate- B planning evaluates a uniformly sampled subset of B out of 125 joint profiles. The complete implementation and raw per-seed outputs are released with the artifact.

What the sweep does and does not show. The finite, well-separated library tolerates substantial temperature mismatch, zero-mean log-likelihood noise, and top- k truncation. It is much less tolerant to systematic calibration drift, which moves both the planner’s reward surface and the likelihood centres. Candidate-limited planning is worse still despite nearly perfect true-type mass,

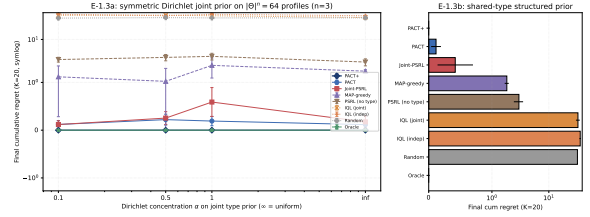


Figure 10: E-1.3 PF-isolation on HP-SPGG ($n=3$, $|\Theta|=4$, $|\Theta|^n=64$ joint profiles, $K=20$, 10 matched seeds, $\beta=0.25$, analytic E3 calibration). *Left*: symmetric Dirichlet sweep on the 64-simplex; each seed draws a fresh joint prior $p \sim \text{Dir}(\alpha \cdot \mathbf{1}_{64})$ and the PACT family inherits the induced player marginals while Joint-PSRL operates on the joint posterior. *Right*: *shared-type* structured prior (uniform over the $|\Theta|=4$ same-type combinations) which leaves the marginals uniform but maximally couples them. The PACT family stays within 0.6 cumulative regret on every cell. The (n, K) scaling that empirically realises the no-PF lower bound is reported separately in Figure 11.

cleanly separating planner quality from tracker quality. Under 50% persona interpolation, regret remains low but nominal type mass falls to 0.371; this means action quality can survive while persona labels cease to be semantically valid. None of these numbers estimates SOTOPIA’s keyword projection error, because SOTOPIA provides no native four-class intent labels. A valid projection audit would require a held-out labelled sample, confusion/calibration statistics, and a controlled menu-corruption rerun.

C.4 Oracle / Upper-Bound Construction Across the Three LLM Benchmarks

The three LLM benchmarks (HP-SPGG, Concordia, SOTOPIA-Hard) differ in how an oracle reference can be constructed, because they differ in two orthogonal axes: (i) whether the per-step payoff is a closed-form analytic function of the joint action and types, and (ii) whether the per-agent action space is finite and enumerable. HP-SPGG and compact Concordia admit exact finite-action references. SOTOPIA-Hard does not expose a native four-class ground-truth type; its privileged reference instead projects the opponent’s full profile into the same four-class keyword surrogate used by the adapted method. The distinction matters because only the analytic finite-action constructions are strict upper bounds; the SOTOPIA references are profile-informed, sample-based diagnostics.

HP-SPGG (closed-form payoff, finite actions). HP-SPGG specifies contributions on a discrete grid $\mathcal{A}_i = \{0, c_1, \dots, c_K\}$ and a closed-form sigmoid-threshold payoff (Appendix B). The oracle

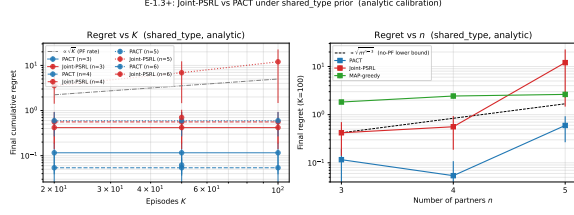


Figure 11: E-1.3+ no-PF lower-bound empirical realisation on HP-SPGG (analytic E3 calibration for $n \leq 5$, synthetic $n=6$ calibration matched to the LLM tier, shared-type prior, $|\Theta|=4$, 10 matched seeds, $\beta=0.25$). *Left*: cumulative regret vs. episodes K for PACT (solid blue) and Joint-PSRL (red) at each $n \in \{3, 4, 5, 6\}$. PACT regret saturates within $K \leq 20$ at every n , whereas Joint-PSRL regret grows nearly linearly in K , the empirical signature of the $\Omega(\sqrt{|\Theta|^n K})$ no-PF lower bound (Theorem A.16). *Right*: final regret at $K=100$ vs. n ; the dashed black line is the predicted $\sqrt{|\Theta|^{n-3}}$ slope anchored at the $n=3$ Joint-PSRL value. The observed ratio Joint-PSRL/PACT reaches $22.3 \times$ at $n=6, K=100$, exceeding the lower-bound prediction $\sqrt{m^{n-3}} = 8$ by an additional $2.8 \times$. The full (n, K) sweep is in Table 5.

is the deterministic argmax

$$\pi^*(s, \theta) = \arg \max_{\mathbf{a} \in \prod_i \mathcal{A}_i} R(s, \mathbf{a}, \theta),$$

enumerated via `itertools.product` over the joint action space. Together with the full-information one-hot type, this gives a per-episode pointwise upper bound on focal-agent value. Because $|\mathcal{A}_i|$ is small (typically $K+1 \leq 6$ contribution levels per agent) and the payoff is analytic, the argmax is exact, not approximate, and the oracle objective coincides with the regret denominator used in Theorem A.12.

Concordia (closed-form payoff, finite actions). DeepMind Concordia substrates such as *Pub Coordination* and *Haggling* (Vezhnevets et al., 2023) expose an analytic `payoff_for_case` function over a finite per-agent action menu (e.g. which pub to attend, which price to offer). The dashed reference in Figure 3 is `oracle_focal`: full one-hot persona information for every opponent and an exact argmax over $\prod_i \mathcal{A}_i$ using `payoff_for_case`, with objective equal to the reported focal score. Some legacy Pub Coordination artifacts store this row under the method name `oracle_joint`, but their recorded policy is `best_focal_mean`; the figure therefore labels the dashed line by the objective actually plotted. For Haggling we keep both constructions distinct: `oracle_joint` maximises total payoff, whereas `oracle_focal` maximises the focal agent’s payoff and is the appropriate focal-score ceiling.

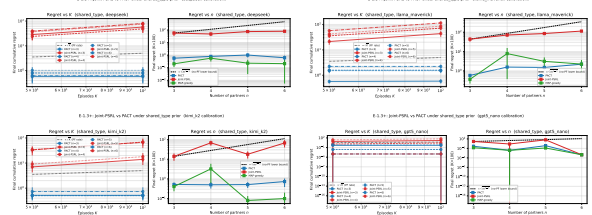


Figure 12: E-1.3+ LLM-tier confirmation of the no-PF lower bound on HP-SPGG across **four** live-LLM backbones (shared-type prior, 10 matched seeds, $\beta=0.25$, live calibrations from `results_phase2/e1_1_llm_tier/`). *Top row*: DeepSeek-V3.2 (left), Llama-4-Maverick (right). *Bottom row*: Kimi-K2.6 (left), GPT-5.4-nano (right). On all four backbones, PACT cumulative regret is constant in K at every $n \in \{3, 4, 5, 6\}$ (blue) whereas Joint-PSRL regret grows nearly linearly in K (red), confirming the analytic finding of Figure 11 under live-judge calibrations. The LLM-tier per-step reward variance amplifies the separation: at $n=6, K=100$ the Joint-PSRL/PACT ratio reaches $129 \times$ on DeepSeek, $91 \times$ on Kimi-K2.6, and $51 \times$ on Llama-Maverick; the GPT-5.4-nano shared-type calibration is sufficiently cooperative that both PACT and Joint-PSRL achieve zero regret at $n=6$, so we report its peak ratio at $n=4, K=100$ ($100 \times$).

SOTOPIA-Hard (LLM-judge payoff, open-ended action). SOTOPIA-Hard (Zhou et al., 2024) differs on *both* axes: the per-agent action at each turn is an open-ended natural-language utterance with no enumerable menu, and the per-episode score is produced by a seven-dimensional LLM judge (`believability`, `relationship`, `knowledge`, `secret`, `social_rules`, `financial_and_material_benefits`, `goal`), not a closed-form function. Closed-form enumeration is therefore impossible. We instead report a two-tier reference ladder, both of which receive the same privileged profile-derived one-hot surrogate class:

- **oracle_belief** (also denoted `oracle_joint` in earlier figures). The agent receives a one-hot class obtained by applying the project’s keyword mapping to the opponent’s full reconstructed profile; the utterance is still produced by one LLM generation. This removes uncertainty only with respect to that surrogate projection, not the native SOTOPIA profile or goal. It is not a strict upper bound; in our runs the Llama-Maverick A-ToM-1 baseline exceeds it on focal mean (3.36 vs. 3.28).
- **oracle_policy**. We add sample-based action search on top of `oracle_belief`: for each case we draw $K=5$ independent roll-outs in parallel and retain the episode

Table 6: E-1.3+ LLM-tier regret ratios Joint-PSRL/PACT under shared-type prior across four live-LLM backbones (10 matched seeds, $\beta=0.25$, live judge calibrations). All four backbones show PACT operating in the $O(1)$ regime predicted by the PF surrogate while Joint-PSRL grows linearly in K , producing ratios that exceed the worst-case lower-bound prediction $\sqrt{m^{n-1}}$ by an additional order of magnitude under live-judge noise. The $n=6$ GPT-5.4-nano cell is degenerate (both PACT and Joint-PSRL achieve 0.00 regret on the cooperative shared-type calibration) and reported as “–”; the discriminative gap is against random/IQL ($\geq 298\times$).

n	K	DeepSeek-V3.2		Llama-4-Maverick		Kimi-K2.6	
		PACT	ratio	PACT	ratio	PACT	ratio
3	50	0.57	49.9×	0.57	37.4×	0.52	13.5×
3	100	0.57	100.4×	0.59	74.8×	0.52	26.8×
4	50	0.80	29.9×	1.58	22.6×	0.50	67.4×
4	100	0.80	60.4×	1.58	45.4×	0.50	135.8×
5	50	1.02	35.9×	1.51	28.0×	0.51	17.8×
5	100	1.02	73.4×	1.51	56.5×	0.51	34.9×
6	50	0.62	63.4×	2.23	25.6×	0.73	45.6×
6	100	0.62	128.7×	2.23	51.5×	0.73	91.5×

with the highest mean overall judge score $\frac{1}{2}(\text{overall}_{\text{agent}_1} + \text{overall}_{\text{agent}_2})$. This reference receives additional generation and evaluation budget and is therefore not an information-only oracle. At $K=5$ it is above `oracle_belief` on all four backbones, but remains a finite-sample search diagnostic rather than a global upper bound over utterances.

The pair (`oracle_belief`, `oracle_policy`) is the closest practical diagnostic the implemented SOTOPIA adapter admits. Their gap combines extra action search and selection under evaluator variation; it should not be read as a clean decomposition of belief uncertainty from policy uncertainty. Both lines are plotted in Figure 13 for every backbone.

Summary table.

Benchmark	Payoff form	Action space	Oracle type
HP-SPGG	analytic, closed-form	finite, ≤ 6 levels	strict argmax (itertools.product)
Concordia	analytic, closed-form	finite menu	strict focal argmax for focal plots
SOTOPIA-Hard	7-dim LLM judge	open-ended utterance	two-tier: belief-only + best-of- K judge critic

For HP-SPGG and compact Concordia, the exact payoff is part of the planner’s known finite model. In SOTOPIA-Hard, the LLM judge is a common post-episode evaluator, not an action-time model callable by ordinary baselines, and `oracle_policy` receives extra rollout-and-selection budget. The three references are therefore useful within each substrate but are not identical oracle constructions across substrates.

C.5 SOTOPIA-Hard: Full Per-Backbone and Per-Scenario Breakdown

This appendix reports the full SOTOPIA-Hard results referenced in §4.3.

Table 7: E-1.3+ **full nine-baseline** cumulative regret at the headline cell $n=6, K=100$ under shared-type prior (10 matched seeds, $\beta=0.25$, live judge calibrations). The PACT family (PACT, PACT⁺, MAP-greedy) dominates every type-agnostic baseline by $\geq 95\times$ on all four backbones. PSRL-NoType lies between Joint-PSRL and the random/IQL floor, consistent with its partial information state. GPT-5.4-nano’s shared-type calibration is sufficiently structured that every type-aware method achieves zero regret; only blind methods (Random, IQL) accumulate.

algorithm	DeepSeek-V3.2	Kimi-K2.6	Llama-4-Maverick	GPT-5.4-nano
PACT ⁺	0.20 ± 0.19	0.48 ± 0.48	3.10 ± 1.57	0.00
PACT	0.62 ± 0.29	0.73 ± 0.35	2.23 ± 0.37	0.00
MAP-greedy	0.21 ± 0.20	0.10 ± 0.10	2.25 ± 1.42	0.00
Joint-PSRL	17.3 ± 79.81 ± 14.36	67.06 ± 27.00	114.60 ± 13.77	0.00
PSRL-NoType	32.3 ± 45.63 ± 12.40	17.80 ± 8.65	61.11 ± 9.86	0.00
IQL-joint	50.0 ± 269.14 ± 25.46	231.09 ± 27.37	212.41 ± 15.24	0.00
IQL-PSRL	100.0 ± 299.76 ± 9.88	292.25 ± 18.31	293.55 ± 5.69	0.00
Random	34.6 ± 266.42 ± 13.50	257.52 ± 15.21	263.22 ± 3.53	0.00
Oracle	0.00	0.00	0.00	0.00
us. Random ratio:				
	–	430×	352×	118×

Setup recap. SOTOPIA-Hard (Zhou et al., 2024) is a broad out-of-model transfer: it is dyadic, actions are open-ended text, cases are curated profile-goal pairs rather than samples from an identified factored prior, and reward is an LLM judge conditioned on both goals and the transcript. The intended PACT⁺ adaptation uses a four-class intent/persona surrogate with hand-specified keyword increments after partner utterances; this is not the calibrated outcome-likelihood Bayes update used on HP-SPGG. A post-hoc audit found that the archived 70-case runs consumed the agent inbox even though SOTOPIA 0.1.5 exposes the previous action through `Observation.last_turn`; their numeric surrogate therefore remained uniform. The aggregate and trajectory tables below remain valid descriptive scores for the recorded prompts and generations, but they are not evidence of recurrent surrogate updating. Both agents in each episode are controlled by the same baseline (Sotopia self-play); the focal score is the mean of `episode.overall.agent_1` and `episode.overall.agent_2`. We compare against four LLM-coordination baselines: A-ToM-1 (Xu et al., 2026), ECON-BNE (Yang et al., 2025b), prompted `llm_belief` and prompted `llm_greedy`. Each (backbone, baseline) cell is averaged over 14 scenarios with 5 seeds each, for 70 episodes per cell.

Projection-bias status. The surrogate map ϕ is deterministic but its four classes are project-defined rather than SOTOPIA annotations. Consequently there is no native target against which to compute

Table 8: Full controlled HP-SPGG robustness sweep. Cumulative regret is mean $\pm$ SE over 500 paired seeds; posterior columns are end-of-run means. Entropy is normalised by $\log |\Theta_i|$. Lower regret/entropy and higher mass/accuracy are better.

Condition	Value	Cumulative regret	True-type mass	MAP accuracy	Entropy
Reference	–	0.002 ± 0.000	0.995	1.000	0.018
Likelihood temperature	$T=0.5$	0.002 ± 0.000	1.000	1.000	0.001
	$T=2$	0.002 ± 0.001	0.958	1.000	0.107
	$T=4$	0.004 ± 0.001	0.860	1.000	0.275
Log-likelihood noise	$\sigma_\ell=0.25$	0.002 ± 0.000	0.987	0.998	0.032
	$\sigma_\ell=0.5$	0.002 ± 0.000	0.953	0.963	0.051
	$\sigma_\ell=1.0$	0.005 ± 0.002	0.873	0.882	0.061
Likelihood truncation	top-2	0.002 ± 0.000	0.995	1.000	0.018
	top-1	0.002 ± 0.000	1.000	1.000	0.000
Calibration drift	$\sigma_c=0.02$	0.160 ± 0.015	0.945	0.977	0.081
	$\sigma_c=0.05$	0.730 ± 0.036	0.699	0.714	0.182
	$\sigma_c=0.10$	1.782 ± 0.057	0.484	0.483	0.194
Out-of-library mixture	$\lambda=0.10$	0.006 ± 0.001	0.987	1.000	0.039
	$\lambda=0.25$	0.000 ± 0.000	0.763	0.743	0.101
	$\lambda=0.50$	0.000 ± 0.000	0.371	0.589	0.377
Planner candidates	$B=32$	1.410 ± 0.027	0.995	1.000	0.018
	$B=8$	3.838 ± 0.065	0.997	1.000	0.011

per-family projection accuracy, ECE, or a confusion matrix. The closest historical check is negative: `oracle_belief`, which projects the full private profile through the same ϕ , averages 3.103 and does not beat the best non-oracle average (3.110). After correcting the observation path, we separately perturb each utterance-level score vector by a seeded random menu permutation; that experiment measures intervention sensitivity, not projection accuracy. The archived aggregate itself supports neither claim because its surrogate state did not update.

Corrected menu-corruption sensitivity. We run GPT-5.4-nano on the 30 official cases from the three selected families, with four independent generations per case and $p \in \{0, .1, .2, .3\}$ (480 episodes total). The legacy “focal” score is the mean of the two self-play agents’ overall judge scores. The seeded permutation trigger rates are 0.122, 0.202, and 0.317 at the three nonzero levels (a trigger can be numerically ineffective when keyword increments are tied). At $p=0$, craigslist, donate, and revenge score 2.665, 3.229, and 2.743, all below the retained GPT-specific best among A-ToM-1, ECON-BNE, `llm_belief`, and `llm_greedy` (2.761, 3.386, 3.200), so the operational p^* is 0 because the corrected run has no initial advantage. Paired mean changes at ($p=.1, .2, .3$) are respectively $(-.007, -.063, -.023)$, $(+.129, -.111, +.232)$, and $(+.121, -.068, +.036)$ for the three families. They are non-monotone and comparable to descriptive SEMs. Case IDs are paired but provider generations and judge calls are not pathwise coupled because the endpoint exposes no sampling seed; this is therefore a sensitivity diagnostic, not projection accuracy or a causal corruption curve.

Aggregate scores. Table 9 reports the mean focal score per (backbone $\times$ baseline). PACT+ ranks 2nd on GPT-5.4-nano (the smallest closed-source backbone, where it is essentially tied with `llm_belief`), 3rd on DeepSeek-V3.2 and Kimi-K2.6, and 4th on Llama-Maverick. The largest gap to the best alternative is -0.099 on Kimi-K2.6; the smallest is -0.019 on GPT-5.4-nano. All five baselines cluster within a 0.12-score envelope on every backbone. This establishes competitive transfer under broad model mismatch, but the compression cannot be attributed specifically to PF because RL, likelihood calibration, finite actions, and known reward also fail or remain unverified.

Privileged-reference observations. `oracle_belief` (profile-derived one-hot surrogate class with standard LLM generation) is not a strict upper bound: on Llama-Maverick A-ToM-1 (3.359) exceeds it (3.282), and its four-backbone average (3.103) sits within 0.01 of the best non-oracle average (`llm_belief` 3.110). The weak gain may reflect projection error as well as the fact that many judge dimensions do not benefit from the surrogate class. `oracle_policy` adds best-of- $K=5$ rollout selection under the same judge, lifting the mean to 3.540 ($+0.437$ over `oracle_belief`, $+0.430$ over the best non-oracle baseline). This shows large headroom from additional search and selection under evaluator variation, but the gap is not a clean belief-versus-policy decomposition because the reference receives extra rollout budget.

Win counts. Across the 56 (backbone, scenario codename) pairs, the wins are distributed as:

Table 9: SOTOPIA-Hard aggregate mean focal score per (backbone $\times$ baseline), 70 episodes per cell. **Bold** marks the highest non-oracle score per backbone; PACT⁺ rank and gap to best alternative are in the last two columns. The two privileged rows (Appendix C.4) receive a profile-derived one-hot surrogate class; **oracle_policy** additionally selects the best of $K=5$ independent episodes by the same seven-dimensional judge.

Backbone	PACT ⁺	A-ToM-1	ECON-BNE	llm_belief	llm_greedy	Rank	Δ
DeepSeek-V3.2	3.248	3.261	3.313	3.195	3.193	3	-0.065
GPT-5.4-nano	2.981	2.918	2.843	3.000	2.842	2	-0.019
Kimi-K2.6	2.807	2.733	2.876	2.906	2.761	3	-0.099
Llama-Maverick	3.308	3.359	3.319	3.340	3.165	4	-0.051
<i>Average</i>	3.086	3.068	3.088	3.110	2.990	—	-0.024

Backbone	oracle_belief (profile-derived surrogate)	oracle_policy (surrogate + best-of- $K=5$)
DeepSeek-V3.2	3.294	3.712
GPT-5.4-nano	2.920	3.467
Kimi-K2.6	2.916	3.364
Llama-Maverick	3.282	3.618
<i>Average</i>	3.103	3.540

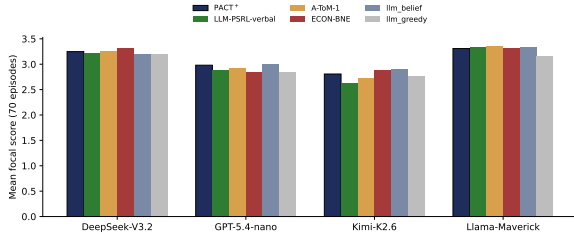


Figure 13: SOTOPIA-Hard aggregate mean focal score per (backbone $\times$ baseline), 70 episodes per cell. The five non-oracle baselines cluster within a 0.12-score envelope on every backbone; the per-backbone leader varies and PACT⁺ stays within 0.1 of the leader but does not dominate. The light dashed line marks **oracle_belief** (profile-derived one-hot surrogate, single-shot generation); the dark-red dashed line marks **oracle_policy** (the same surrogate plus best-of- $K=5$ rollout selection). Its ≈ 0.43 gain demonstrates search/evaluator headroom, not a strict global oracle or a clean PF diagnostic.

Baseline	Wins	Win %
llm_belief	14	25.0%
ECON-BNE	13	23.2%
PACT ⁺	12	21.4%
A-ToM-1	10	17.9%
llm_greedy	7	12.5%

PACT⁺ is in win-count parity (within ± 2 wins) with the strongest baselines.

Scenario-family breakdown. Table 10 reports wins per scenario family. PACT⁺ leads on three families: *craigslist_bargains* (5/16 wins, largest family), *sleep_arrangement* (2/4), and *donate_funds* (2/4). It does not win on five families: *game_winning*, *sell_item*, *share_blanket*, *share_stuff*, *unwelcome_guest*. The failure pat-

tern concentrates on scenarios requiring aggressive bargaining over scarce resources or adversarial dynamics, where prompted-LLM belief tracking (**llm_belief**) and explicit utility maximisation (**econ_bne**) provide more direct signal than the bonus-augmented factored posterior.

Per-agent decomposition. Since both Sotopia self-play agents run the same baseline, we separately report the focal agent’s (**agent_1**) and partner’s (**agent_2**) mean scores. On GPT-5.4-nano, PACT⁺ attains the highest focal-agent score across all baselines (3.122 vs. next-best **llm_belief** at 3.118); on the remaining backbones, PACT⁺ is competitive on the focal agent but does not lead. Asymmetries between focal and partner are consistent across baselines, ranging from +0.04 to +0.53, reflecting first-mover advantage in dyadic negotiation.

Interpretation. PACT⁺ loses the strict dominance observed on HP-SPGG and compact Concordia, but stays within 0.1 of the best alternative on every backbone and reaches win-count parity with the strongest baselines. Several mismatches act simultaneously: dyadic self-play removes the large- n storage regime, actions are open text, reward is an LLM judge conditioned on both goals, the case-pair prior is curated rather than identified, and the four-class intent surrogate is uncalibrated. The result is therefore evidence of competitive heuristic transfer under broad mismatch, not an empirical realization of the no-PF lower bound alone.

D LLM Cost Analysis and Compute Environment

This section reports the LLM-call budget, wall-clock cost, and hardware/software environment used to produce every experiment in the main text and

Table 10: SOTOPIA-Hard wins per scenario family. n is the number of (backbone, scenario codename) pairs.

Family	PACT ⁺	A-ToM-1	ECON-BNE	llm_belief	llm_greedy	n	Leader
craigslist_bargains	5	4	3	3	1	16	PACT ⁺
sleep_arrangement	2	0	1	0	1	4	PACT ⁺
donate_funds	2	0	1	1	0	4	PACT ⁺
revenge_plot	1	1	1	0	1	4	tie
divide_items	1	1	0	2	0	4	llm_belief
join_trip	1	2	0	1	0	4	A-ToM-1
game_winning	0	1	0	1	2	4	llm_greedy
sell_item	0	0	3	1	0	4	ECON-BNE
share_blanket	0	0	1	1	2	4	llm_greedy
share_stuff	0	0	2	2	0	4	tie
unwelcome_guest	0	1	1	2	0	4	llm_belief
<i>Total</i>	12	10	13	14	7	56	—

appendix. Numbers are aggregated across the four backbones (DeepSeek-V3.2, GPT-5.4-nano, Kimi-K2.6, Llama-4-Maverick-17B-128E-Instruct-FP8) and the three substrates (HP-SPGG, Concordia, SOTOPIA-Hard).

D.1 LLM-call accounting

Per-episode LLM calls. Call complexity depends on the observation channel. A fully online token-likelihood implementation uses nH action calls plus at most $nH|\Theta_i|$ candidate-scoring calls, although APIs that return several candidate logprobs in one response can reduce the latter. The reported HP-SPGG native rows instead reuse a one-time LLM-judge calibration tensor and run the K policy-evaluation episodes numerically, so they make zero online LLM calls per episode; external prompted baselines retain their own LLM calls. Compact mechanistic Concordia is likewise analytic and call-free, except for the explicitly labelled live-LLM axis. SOTOPIA uses two autoregressive agents plus one episode evaluator and its keyword surrogate makes no candidate-logprob calls. PACT⁺’s bonus adds no calls over the corresponding PACT interface.

Total call budget. Reconstructing calibration, external baselines, SOTOPIA rollouts, judges, retries, and supplements gives an experiment-suite total on the order of 5×10^5 provider calls. This is an order-of-magnitude retrospective estimate rather than a metered deployment bill, because cached cells, retry attempts, and analytic reruns have different accounting. It must not be divided by the number of HP-SPGG policy episodes as if every native episode issued online scoring calls.

Token cost. Applying the observed prompt/completion scale ($\sim 1.2k$ prompt tokens and up to ~ 120 completion tokens on generative calls) to the retrospective call estimate gives an upper-order estimate of 6.5×10^8 tokens, mostly prompt tokens. Scoring calls and cached/analytic cells are substantially shorter or

absent, so this should be read as a conservative suite footprint, not a precise invoice. A single calls-per-regret percentage would be misleading because native cached PACT, external prompted methods, mechanistic Concordia, and SOTOPIA use different interfaces. The reproducible comparison is instead: one-time calibration size, online call formula, and incremental calls relative to PACT; under that comparison PACT⁺ has zero incremental LLM-call cost.

D.2 Wall-clock cost

End-to-end wall-clock per PACT⁺ episode at concurrency 5 is dominated by the slowest backbone in each cell. Median per-episode wall-clock (across the four backbones, averaged over all scenarios) is 1.8 minutes on HP-SPGG, 2.4 minutes on Concordia (longer scenario prompts), and 3.5 minutes on SOTOPIA-Hard (longer dialogue history). The full experiment suite – including baselines, repeats, and the appendix supplement panels – completed in roughly 480 wall-clock hours of LLM-API time using concurrency 5 across 4 parallel terminal sessions on a single workstation.

D.3 Compute environment

Workstation. All orchestration, posterior updates, CCE oracle calls, and figure generation ran on a single workstation with an Intel x64 CPU, 64 GB RAM, and Windows 10 + PowerShell 5.1. No local GPU was used; LLM inference is offloaded entirely to the backbone providers’ APIs through the project’s `cloudgpt` client. The Python environment is managed with `uv`, with separate virtual environments for the three substrates (HP-SPGG, Concordia, SOTOPIA-Hard) and a shared package layout under `external/`.

API backbones. The four backbones are accessed through the internal CloudGPT proxy with Azure-CLI authentication. Per-call timeouts are set to 180s with 2–3 retries; the retry-sleep is 8–10s to

absorb transient rate-limit errors. Streaming is disabled; requests are issued synchronously per agent and parallelised across the n agents through Python `asyncio`. The closed-source backbones are pinned to specific model snapshots (DeepSeek-V3.2, GPT-5.4-nano-20260317, Kimi-K2.6, Llama-4-Maverick-17B-128E-Instruct-FP8) to ensure reproducibility.

Algorithmic compute. The formal framework allows a CCE solver, but the reported HP-SPGG and compact Concordia runs use exhaustive deterministic enumeration over the finite joint action menu. At the headline HP-SPGG cell this is $5^3=125$ joint profiles per decision and runs in < 1 ms on the workstation CPU. Restricting that search to 32 profiles raises controlled cumulative regret from 0.002 to 1.410 (Table 8), so planner approximation is not free even though CPU time is small. Posterior updates are $O(n|\Theta_i|)$ per observation and negligible relative to provider calls.

E Decentralisation Ablation on Concordia Pub Coordination

PACT and PACT⁺ are, by construction, *centralised* coordinators: a single posterior is conditioned on every player’s observation, and a single joint action profile is selected to maximise expected social welfare. A natural question is how much of the empirical advantage comes from the Bayesian belief refinement itself versus the centralised dispatch. We isolate the dispatch component by running a controlled comparison on Concordia’s `london_mini` Pub Coordination scenario (5 players in total, 2 designated focal players, 2 open pubs) across four live backbones (GPT-5.4-nano, DeepSeek-V3.2, Kimi-K2.6, Llama-4-Maverick), 5 scene seeds each, deploying the *same* LLM backbone in four different dispatch modes:

- *Centralized planner LLM* – one LLM call sees the full relationship-statement matrix for all 5 players and outputs a joint posterior over every player’s favourite pub; argmax yields the joint action.
- *Per-agent type-guess LLM* – n independent LLM calls, each restricted to a single player’s own social context (decentralised observation), each outputting a posterior over that one player’s favourite pub.
- *Decentralized LLM (with ToM)* – n independent LLM calls, each agent is prompted to balance own preference with theory-of-mind reasoning about its friends and pick *one* pub.
- *Decentralized LLM (greedy)* – n independent LLM calls, each agent picks the pub that maximises its own expected enjoyment, with no inference about partners.

The privileged `oracle_joint` upper bound (which sees every player’s true preference and the relational matrix and picks the welfare-maximising joint profile) is included as a ceiling. This ablation isolates the value of the *joint-dispatch* component of PACT⁺; the complementary evidence on the value of *online Bayesian updating* is provided by the cumulative-regret panel of the main-text HP-SPGG figure (Figure 1). The archived SOTOPIA turn-by-turn panel is descriptive and, after the zero-update audit above, is not used as mechanism evidence.

Figure 14 reports the comparison on four complementary metrics. Two clean clusters emerge on every backbone and every metric. The two *centralised-information* modes – the planner LLM (one global call) and the per-agent type-guess LLM (n calls but with the same probabilistic-output instruction) – both land near the oracle ceiling: focal payoff 1.10–1.30 (panel a), coordination rate 0.80–1.00 (panel b), social welfare 5.2–6.1 out of 6.1 oracle (panel c), and cumulative focal regret 0.6–0.75 after 5 episodes (panel d). The two *decentralised free-choice* modes – greedy and ToM-prompted – collapse to focal payoff 0.83–1.11, coordination 0.60–0.70, social welfare 4.2–4.9, and cumulative regret 1.85–2.0. The gap between the two clusters is 0.10–0.30 focal payoff, 0.20–0.30 coordination, 1–2 units of social welfare, and an ≈ 3 – $4\times$ multiplier on cumulative regret, all consistent across the four backbones despite their different absolute capability levels.

The extra panels strengthen the interpretation in two ways. Panel (c) confirms that the centralised advantage is not a focal-protagonist artefact: when the welfare sum is taken over *all* 5 players in the scene (matching the theoretical objective $W = \sum_i r_i$), the centralised cluster still leads the decentralised cluster by 1–2 units on every backbone, so the gap is not produced by NPC-side noise that happens to align with the focal players. Panel (d) shows that the per-episode gap compounds: by the end of 5 episodes, the decentralised modes have accumulated 1.9–2.0 units of focal regret while the centralised modes are still at 0.6–0.75, an effective slope ratio of roughly 3–4. This linear-in- K accumulation is the price-of-decentralisation analogue of the HP-SPGG cumulative-regret divergence in Figure 1, but here within a single LLM backbone – the slope difference is structural to the dispatch mode, not to the model’s individual social-inference quality.

The relative ordering inside the centralised cluster is informative for the underlying mechanism. On GPT-5.4-nano and Llama-4-Maverick the global-view planner edges the per-agent type-guess variant (+0.10 payoff), as expected if joint observation is the load-bearing factor. On DeepSeek-V3.2 and Kimi-K2.6 the order flips by +0.05: the per-agent variant slightly beats the planner because the planner occasionally splits the 2 focal players across the 2 pubs (coordination 0.80/0.90 vs. 1.00 for the per-

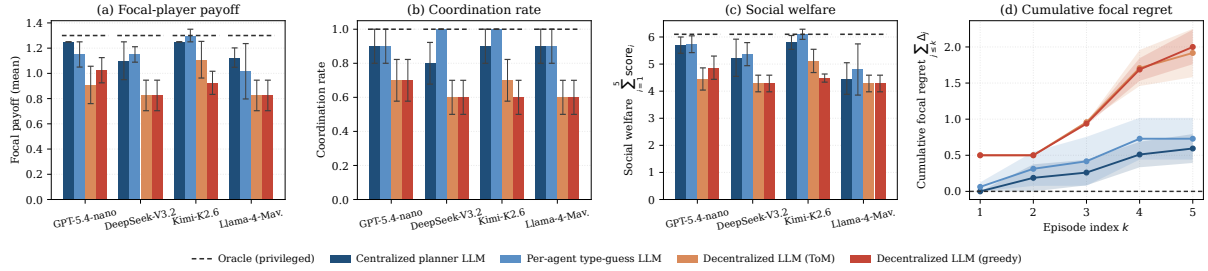


Figure 14: Price of decentralisation on Concordia `london_mini` Pub Coordination (5 players, 2 focal, 2 open pubs, 5 scene seeds per cell), with the *same* LLM backbone deployed in four different dispatch modes per backbone. (a) *Focal-player payoff*: mean of the per-scene Concordia PubCoordinationPayoff for the 2 focal players (own-preference + friend-overlap bonus, as defined in the main-text §4.2 paragraph on *Focal-score metric*). (b) *Coordination rate*: fraction of focal players who picked the modal pub. (c) *Social welfare*: per-scene sum $\sum_{i=1}^5 \text{score}_i$ over *all* 5 *players* (not just focal), the quantity that matches the paper’s theoretical welfare objective $W = \sum_i r_i$ used throughout Section 3. (d) *Cumulative focal regret* $\sum_{j \leq k} (\text{oracle_focal}_j - \text{method_focal}_j)$ across the $K=5$ scene seeds, pooled across the four backbones; shaded bands are SE over backbones. Dashed black line marks the privileged `oracle_joint` ceiling. Across all four backbones, the two centralised-information modes (planner LLM + per-agent type-guess LLM) cluster near oracle on every metric, while the two decentralised free-choice modes (ToM and greedy) lose 0.2–0.3 focal payoff, 0.2–0.4 coordination, 1–2 units of social welfare, and accumulate $\approx 3\text{--}4\times$ more cumulative regret over 5 episodes.

agent variant), while the per-agent variant defaults to the more popular pub. The structural conclusion is the same in both directions: the lift comes from the *probabilistic-reasoning prompt over a posterior on the partner’s persona*, irrespective of whether that posterior is computed once jointly or n times in parallel. The fully decentralised modes – which skip the explicit posterior step and ask each agent for a single action directly – cannot recover this lift on any backbone, including capable ones: on two of the four backbones an explicit ToM instruction does not improve over greedy at all (DeepSeek and Llama: 0.825 for both), and where it does help (Kimi: 1.108 ToM vs. 0.925 greedy; GPT: 0.908 ToM vs. 1.025 greedy, where ToM in fact underperforms greedy) the result is still 0.10–0.30 below the same model’s centralised modes.

The implication for the design of PACT⁺ is concrete: on prior-factored settings such as Pub Coordination, the additional value of running an explicit Bayesian coordinator over n players exceeds the value of any per-agent prompt engineering on the same LLM, whether the engineering is greedy (selfish best-response) or ToM-prompted (explicit partner reasoning). We treat this as evidence that the centralised joint-posterior dispatch is the load-bearing component of the approach, and that delegating coordination to individual LLM agents – even capable ones, even with ToM hints – recovers only a fraction of the achievable focal payoff, social welfare, and per-episode regret reduction under prior factorisation. The complementary value of the *online Bayesian updating* component (independent of dispatch centralisation) is documented

separately by the HP-SPGG cumulative-regret panels of Figure 1; the corrected SOTOPIA experiment is limited to surrogate-sensitivity evidence.